

Assignment 6 extra

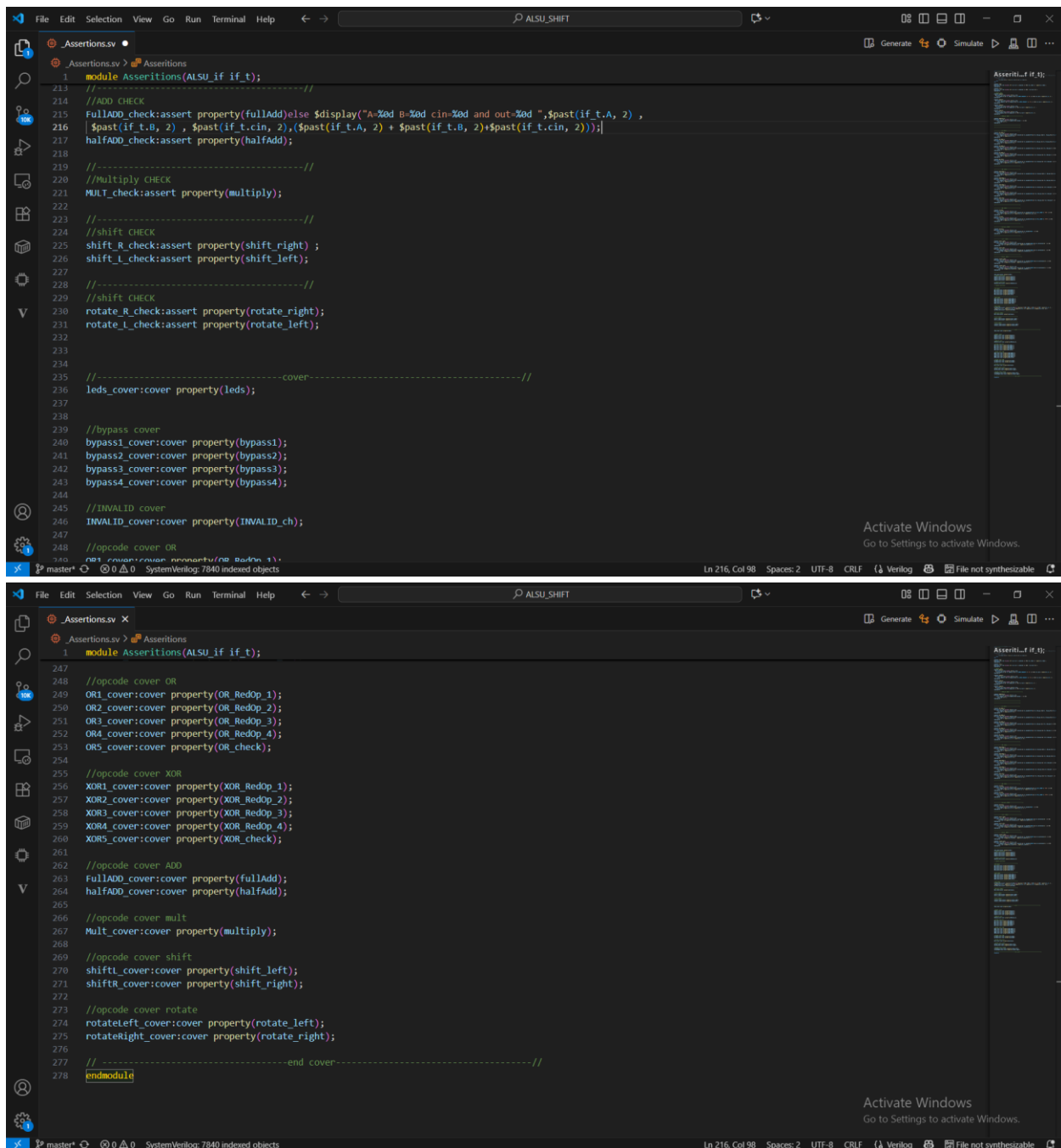
Hadeer Ghoneim

Verification Diploma

1. Assertions


```
File Edit Selection View Go Run Terminal Help  ← →  ALSU_SHIFT  Generate Simulate

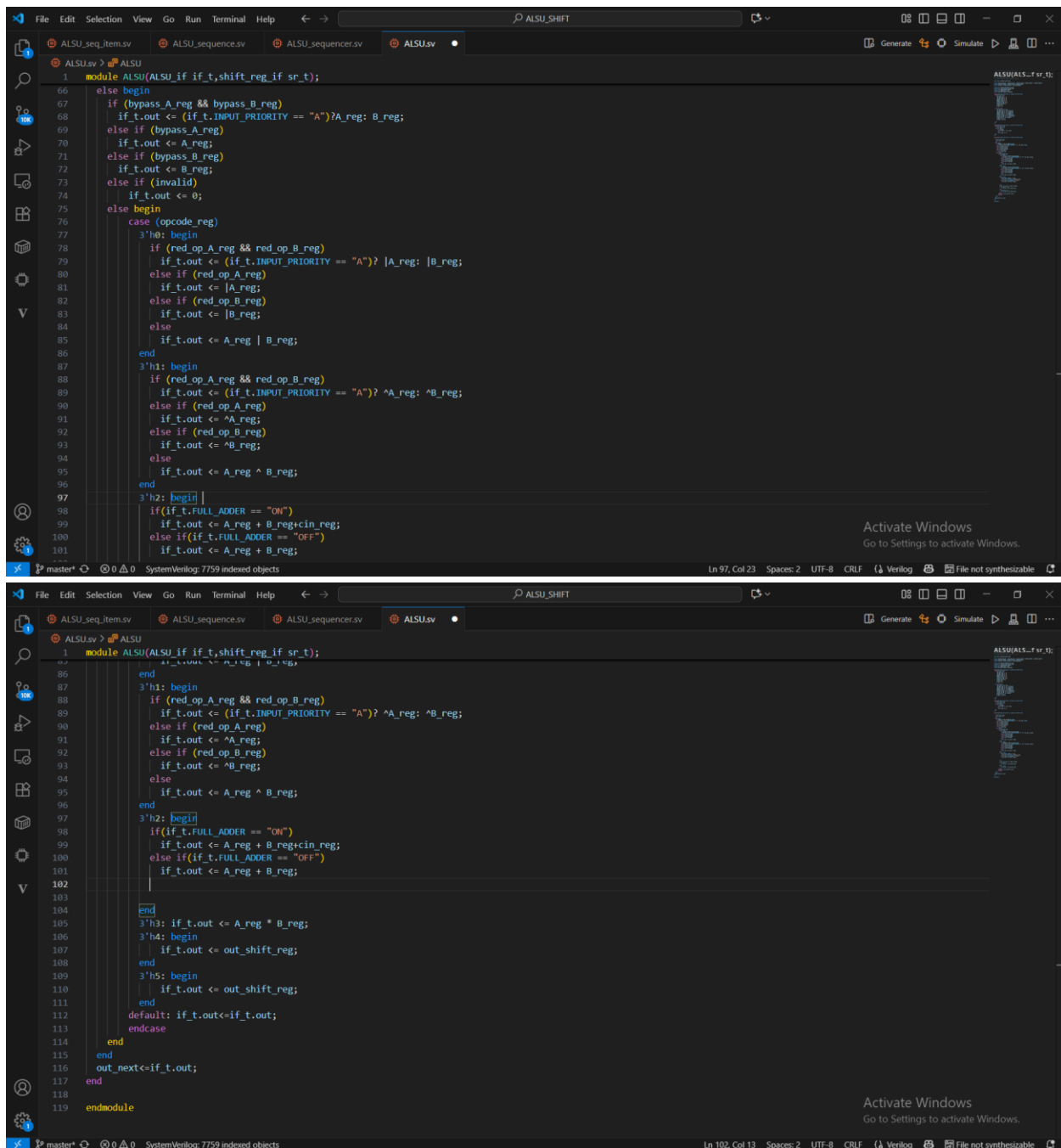
.Assertions.vv x
.Assertions.vv > Assertions > leds
1 module Assertions(ALSU_if if_t);
68
69 property OR_RedOp_3;
70 @ (posedge if_t.clk) disable iff(if_t.rst)
71 ((if_t.bypass_A && if_t.bypass_B && !invalid && if_t.opcode==OR && if_t.red_op_A && !if_t.red_op_B) |> ##1
72 (if_t.out == $past(if_t.A, 2)));
73 endproperty
74
75 property OR_RedOp_4;
76 @ (posedge if_t.clk) disable iff(if_t.rst)
77 ((if_t.bypass_A && if_t.bypass_B && !invalid && if_t.opcode==OR && if_t.red_op_A && if_t.red_op_B) |> ##1
78 (if_t.out == $past(if_t.B, 2)));
79 endproperty
80
81 property OR_check;
82 @ (posedge if_t.clk) disable iff(if_t.rst)
83 ((if_t.bypass_A && if_t.bypass_B && !invalid && if_t.opcode==OR && if_t.red_op_A && if_t.red_op_B) |> ##1
84 (if_t.out == $past(if_t.A, 2) | $past(if_t.B, 2)));
85 endproperty
86
87 //-----
88
89 //-----XOR CHECK-----
90
91 property XOR_RedOp_1;
92 @ (posedge if_t.clk) disable iff(if_t.rst)
93 ((if_t.bypass_A && if_t.bypass_B && !invalid && if_t.opcode==XOR && if_t.red_op_A && if_t.red_op_B && if_t.INPUT_PRIORITY == "A") |> ##1
94 (if_t.out == ^$past(if_t.A, 2)));
95 endproperty
96
97 property XOR_RedOp_2;
98 @ (posedge if_t.clk) disable iff(if_t.rst)
99 ((if_t.bypass_A && if_t.bypass_B && !invalid && if_t.opcode==XOR && if_t.red_op_A && if_t.red_op_B && if_t.INPUT_PRIORITY == "B") |> ##1
100 (if_t.out == ^$past(if_t.B, 2)));
101 endproperty
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```

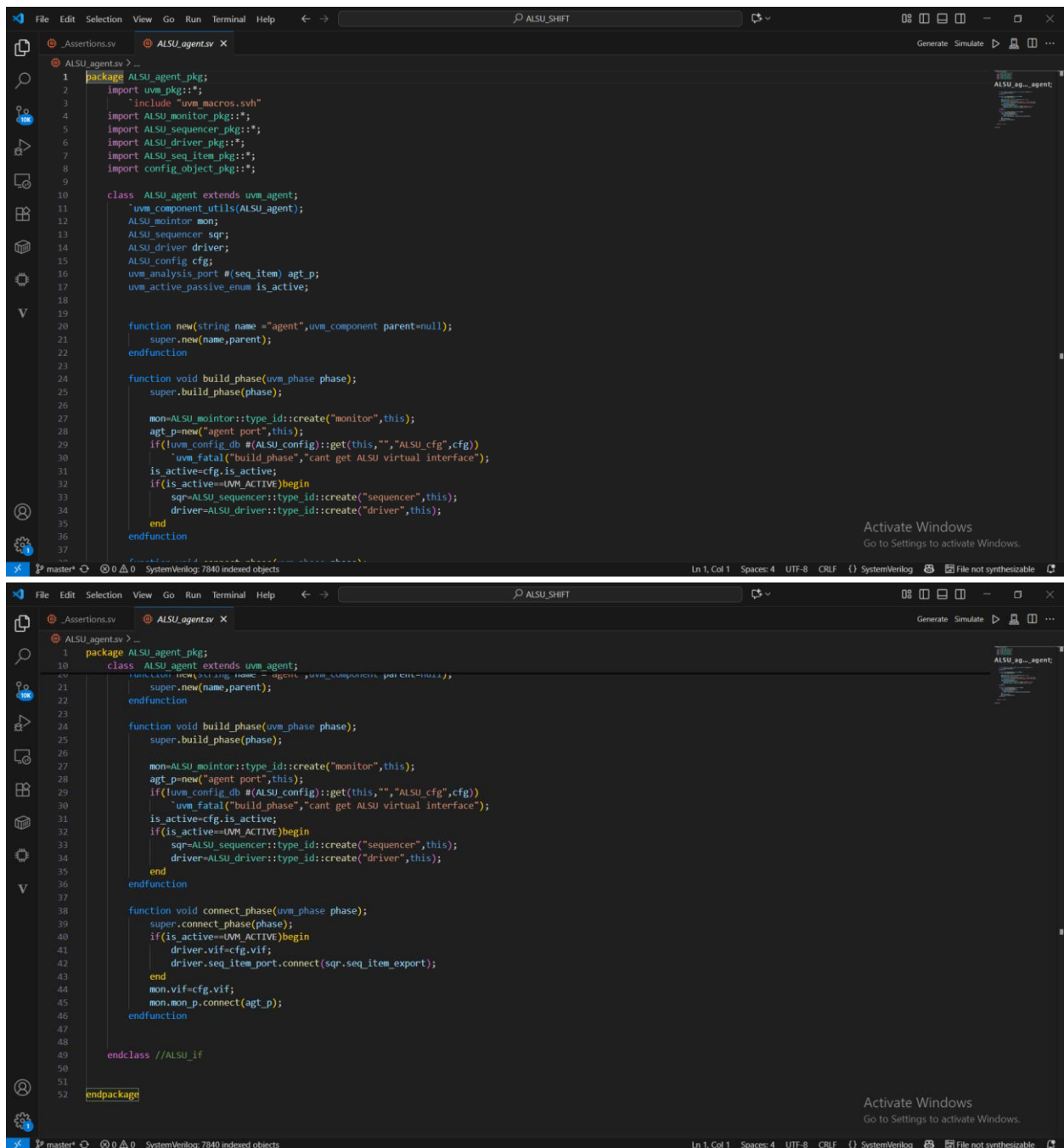
2. ALSU design

```
File Edit Selection View Go Run Terminal Help ALSU_SHIFT
ALSU_seq_item.sv ALSU_sequence.sv ALSU_sequencer.sv ALSU.sv x
Generate Simulate
ALSU
1 module ALSU(ALSU_if if_t, shift_reg_if sr_t);
2
3 reg red_op_A_reg, red_op_B_reg, bypass_A_reg, bypass_B_reg, direction_reg, serial_in_reg;
4 reg signed [1:0] cin_reg;
5 reg [2:0] opcode_reg;
6 reg signed [2:0] A_reg, B_reg;
7 reg signed [5:0] out_next;
8 wire invalid_red_op, invalid_opcode, invalid;
9
10 reg signed [5:0] out_shift_reg;
11
12 assign invalid_red_op = (red_op_A_reg | red_op_B_reg) & (opcode_reg[1] | opcode_reg[2]);
13 assign invalid_opcode = opcode_reg[1] & opcode_reg[2];
14 assign invalid = invalid_red_op | invalid_opcode;
15
16 assign sr_t.serial_in = serial_in_reg;
17 assign sr_t.direction = direction_reg;
18 assign sr_t.mode = opcode_reg[0];
19 assign sr_t.data_in = if_t.out;
20 assign out_shift_reg = sr_t.data_out;
21
22 always @(posedge if_t.clk or posedge if_t.rst) begin
23     if(if_t.rst) begin
24         cin_reg <= 0;
25         red_op_B_reg <= 0;
26         red_op_A_reg <= 0;
27         bypass_B_reg <= 0;
28         bypass_A_reg <= 0;
29         direction_reg <= 0;
30         serial_in_reg <= 0;
31         opcode_reg <= 0;
32         A_reg <= 0;
33         B_reg <= 0;
34     end
35     else begin
36         cin_reg <= if_t.cin;
37         red_op_B_reg <= if_t.red_op_B;
38         red_op_A_reg <= if_t.red_op_A;
39         bypass_B_reg <= if_t.bypass_B;
40         bypass_A_reg <= if_t.bypass_A;
41         direction_reg <= if_t.direction;
42         serial_in_reg <= if_t.serial_in;
43         opcode_reg <= if_t.opcode;
44         A_reg <= if_t.A;
45         B_reg <= if_t.B;
46     end
47 end
48
49 always @(posedge if_t.clk or posedge if_t.rst) begin
50     if(if_t.rst) begin
51         if_t.leds <= 0;
52     end
53     else begin
54         if (invalid)
55             if_t.leds <= ~if_t.leds;
56         else
57             if_t.leds <= 0;
58     end
59 end
60
61 always @(posedge if_t.clk or posedge if_t.rst) begin
62     if(if_t.rst) begin
63         if_t.out <= 0;
64     end
65     else begin
66         if (bypass_A_reg && bypass_B_reg)
67             if_t.out <= (if_t.INPUT_PRIORITY == "A") ? A_reg : B_reg;
68         else if (bypass_A_reg)
69             if_t.out <= A_reg;
70         else if (bypass_B_reg)
71             if_t.out <= B_reg;
72         else
73             if_t.out <= out_next;
74     end
75 end
76 endmodule
```

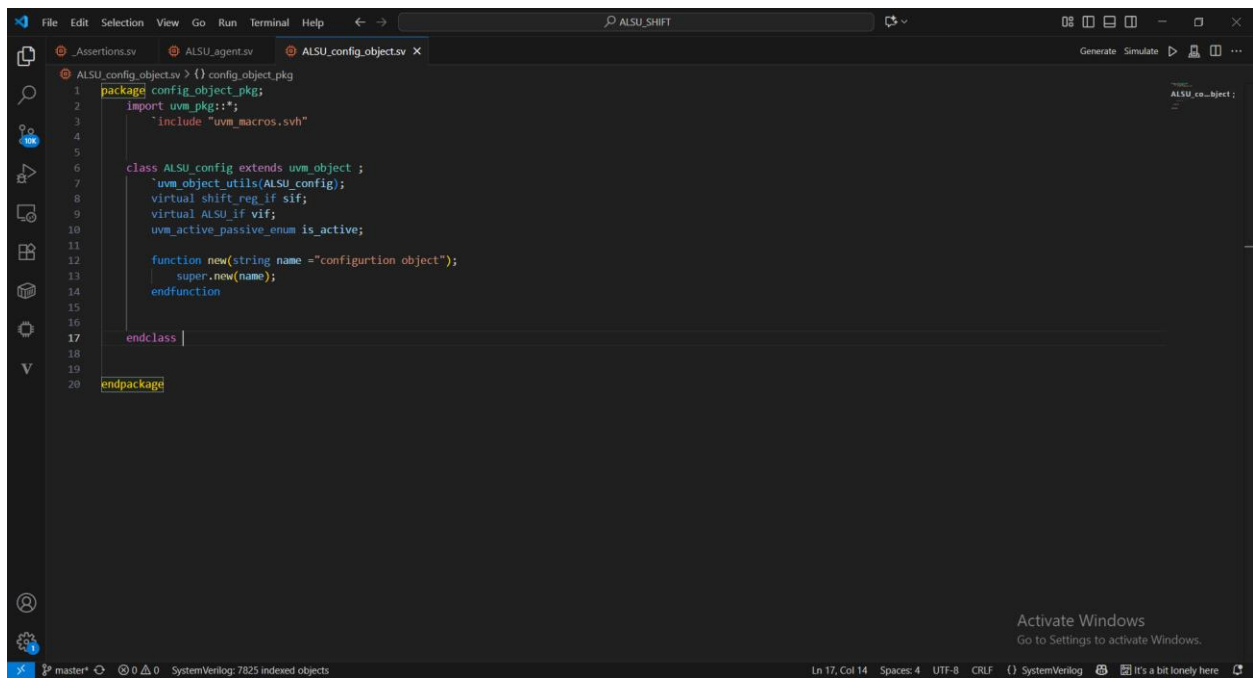
```
File Edit Selection View Go Run Terminal Help ALSU_SHIFT
ALSU_seq_item.sv ALSU_sequence.sv ALSU_sequencer.sv ALSU.sv
Generate Simulate
ALSU
35 end else begin
36     cin_reg <= if_t.cin;
37     red_op_B_reg <= if_t.red_op_B;
38     red_op_A_reg <= if_t.red_op_A;
39     bypass_B_reg <= if_t.bypass_B;
40     bypass_A_reg <= if_t.bypass_A;
41     direction_reg <= if_t.direction;
42     serial_in_reg <= if_t.serial_in;
43     opcode_reg <= if_t.opcode;
44     A_reg <= if_t.A;
45     B_reg <= if_t.B;
46 end
47 end
48
49 always @(posedge if_t.clk or posedge if_t.rst) begin
50     if(if_t.rst) begin
51         if_t.leds <= 0;
52     end
53     else begin
54         if (invalid)
55             if_t.leds <= ~if_t.leds;
56         else
57             if_t.leds <= 0;
58     end
59 end
60
61 always @(posedge if_t.clk or posedge if_t.rst) begin
62     if(if_t.rst) begin
63         if_t.out <= 0;
64     end
65     else begin
66         if (bypass_A_reg && bypass_B_reg)
67             if_t.out <= (if_t.INPUT_PRIORITY == "A") ? A_reg : B_reg;
68         else if (bypass_A_reg)
69             if_t.out <= A_reg;
70         else if (bypass_B_reg)
71             if_t.out <= B_reg;
72         else
73             if_t.out <= out_next;
74     end
75 end
76 endmodule
```



3. ALSU_agent



4. ALSU_config_object



5. ALSU_coverage

The screenshot shows the VS Code editor with the file `ALSU_coverage.v` open. The code is as follows:

```

1 package ALSU_coverage_pkg;
2 import uvm_pkg::*;
3 `include "uvm_macros.svh"
4 import ALSU_seq_item_pkg::*;
5 import constant_enums::*;
6
7 class ALSU_coverage extends uvm_component;
8     `uvm_component_utils(ALSU_coverage);
9
10    seq_item item;
11    uvm_analysis_export #(seq_item) cov_ep;
12    uvm_tlm_analysis_fifo #(seq_item) cov_fifo;
13
14    covergroup cvr_gp;
15
16    CIN:coverpoint item.cin(
17        bins CIN0={0};
18        bins CIN1={1};
19        option.weight=0;
20    )
21
22    red_op_A:coverpoint item.red_op_A(
23        bins red_op_A0={0};
24        bins red_op_A1={1};
25        option.weight=0;
26    )
27
28    red_op_B:coverpoint item.red_op_B(
29        bins red_op_B0={0};
30        bins red_op_B1={1};
31        option.weight=0;
32    )
33
34    bypass_A:coverpoint item.bypass_A(
35        bins bypass_A0={0};
36        bins bypass_A1={1};
37        option.weight=0;
38    )
39
40    bypass_B:coverpoint item.bypass_B(

```

The file explorer on the left shows the project structure with files like `_Assertions.v`, `ALSU_agent.v`, `ALSU_config_objects.v`, and `ALSU_coverage.v`. The output window on the right shows the compilation process with messages like `ALSU_coverage.v:1: package ALSU_coverage_pkg;` and `ALSU_coverage.v:2: import uvm_pkg::*;`.

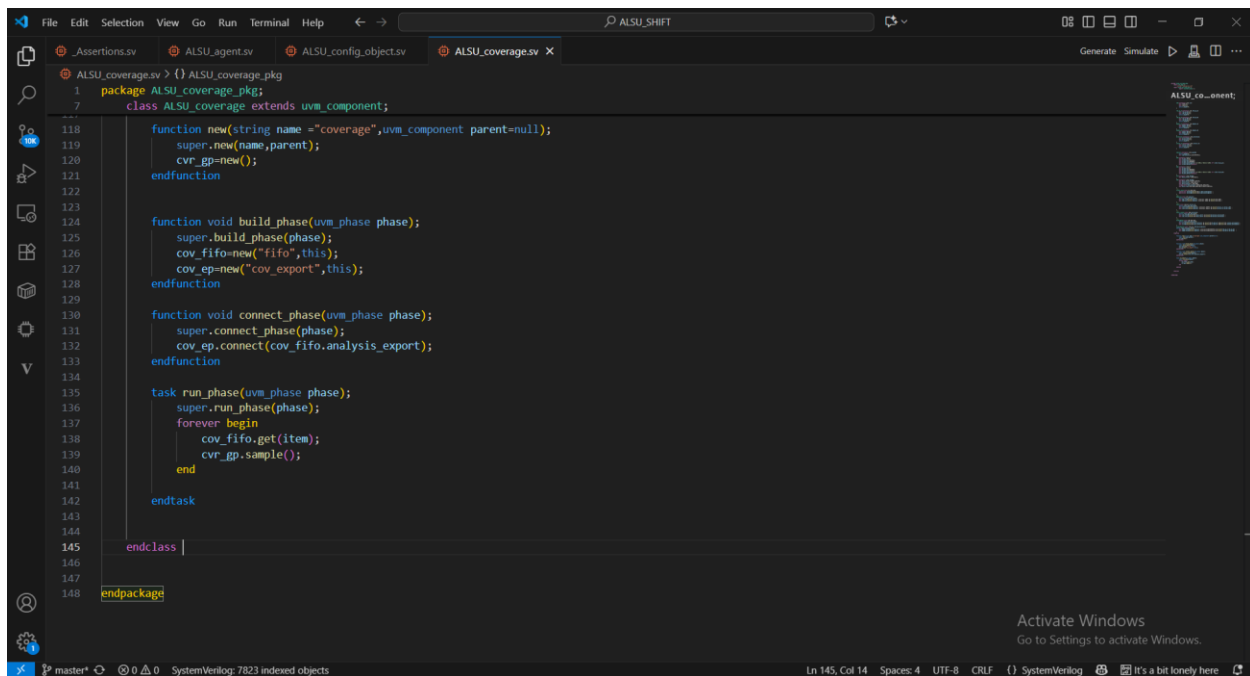
The image shows a screenshot of a SystemVerilog code editor. The main window displays the file `ALSU_coverage.v` with the following code:

```
1 package ALSU_coverage_pkg;
2
3 class ALSU_coverage extends uvm_component;
4
5     bypass_B:coverpoint item.bypass_B{
6         bins bypass_B0={0};
7         bins bypass_B1={1};
8         option.weight=0;
9     }
10
11     direction:coverpoint item.direction{
12         bins direction0={0};
13         bins direction1={1};
14         option.weight=0;
15     }
16
17     serial_in:coverpoint item.serial_in{
18         bins serial_in0={0};
19         bins serial_in1={1};
20         option.weight=0;
21     }
22
23     Special:coverpoint item.opcode{
24         option.weight=0;
25         bins operations[]={OR:ROTATE};
26     }
27
28     CB1:coverpoint item.A{
29         bins A_data_0={0};
30         bins A_data_max={MAXPOS};
31         bins A_data_min={MAXNEG};
32         bins A_data_walkingones[] = {3'b001,3'b010,3'b100} iff (item.red_op_A);
33         bins A_data_default=default;
34     }
35
36     CB2:coverpoint item.B{
37         bins B_data_0={0};
38         bins B_data_max={MAXPOS};
39         bins B_data_min={MAXNEG};
40         bins B_data_walkingones[] = {3'b001,3'b010,3'b100} iff (item.red_op_B);
41         bins B_data_default=default;
42     }
43
44 endclass
```

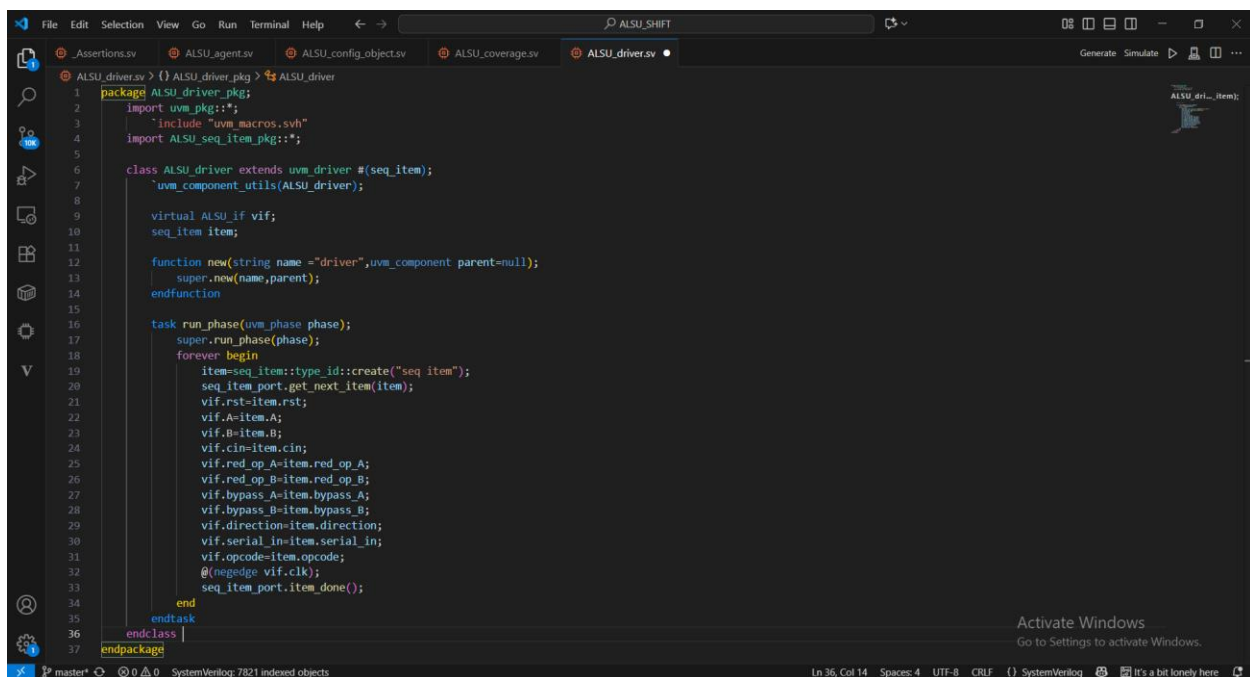
The editor interface includes a top menu bar with File, Edit, Selection, View, Run, Run Terminal, Help, and a search icon. Below the menu is a toolbar with icons for file operations and a search icon. The left sidebar contains a project tree showing the file structure. The right sidebar shows a list of variables and their values. The bottom status bar displays the current line and column (Ln 145, Col 14), the file encoding (UTF-8), and the line endings (CRLF). An "Activate Windows" watermark is visible in the bottom right corner.

```
File Edit Selection View Go Run Terminal Help ALSU_SHIFT Generate Simulate
ALSU_coverage.v > {} ALSU_coverage_pkg
1 package ALSU_coverage_pkg;
7 class ALSU_coverage extends uvm_component;
72 A_Mi:coverpoint item.opcode{
73     bins Bins_arith[] = (ADD,MULT);
74 }
75 CB3:coverpoint item.opcode{
76     bins Bins_shift[] = (SHIFT,ROTATE);
77     bins Bins_arith[] = (ADD,MULT);
78     bins Bins_bitwise[] = (OR,XOR);
79     illegal bins Bins_invalid = (INVALID6,INVALID7);
80     bins Bins_trans = (OR->XOR->ADD->MULT->SHIFT->ROTATE);
81 }
82 //1
83 corner_case:cross A_M,CB2,CB1{
84     ignore_bins walkingA=binsof(CB1.A_data_walkingones) ;
85     ignore_bins walkingB=binsof(CB2.B_data_walkingones) ;
86 }
87 //2
88 Addition:cross CIN,special{
89     option.cross_auto_bin_max=0;
90     bins Add_cin0=binsof(special) intersect (ADD) && binsof(CIN.CIN0) ;
91     bins Add_cin1=binsof(special) intersect (ADD) && binsof(CIN.CIN1) ;
92 }
93 //3
94 shift:cross special,serial_in{
95     option.cross_auto_bin_max=0;
96     bins shift_sio=binsof(special) intersect (SHIFT) && binsof(serial_in.serial_in0) ;
97     bins shift_sii=binsof(special) intersect (SHIFT) && binsof(serial_in.serial_in0) ;
98 }
99 //4
100 shift_rotate:cross CB3,direction{
101     option.cross_auto_bin_max=0;
102     bins shu_rot_d0=binsof(CB3.Bins_shift) && binsof(direction.direction0) ;
103     bins shu_rot_d1=binsof(CB3.Bins_shift) && binsof(direction.direction1) ;
104 }
105 //5
106 walkingones:cross CB3,CB2,CB1{
    ...
}
}
endclass
endpackage
```

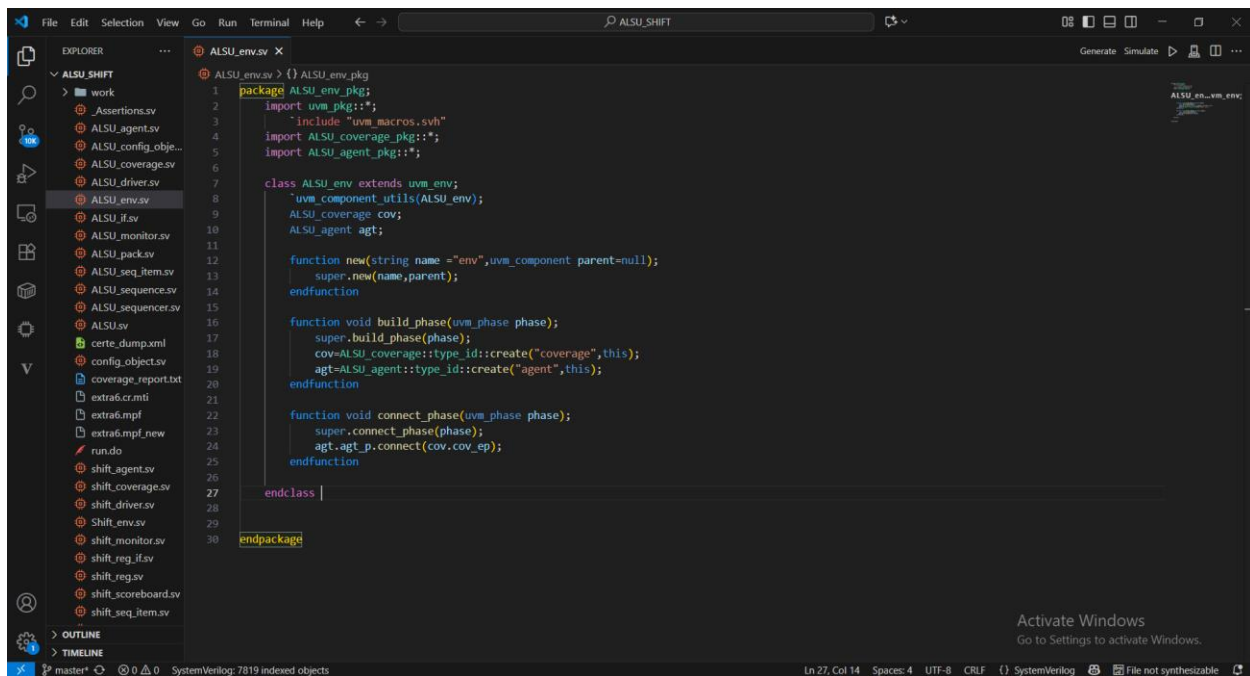
```
File Edit Selection View Go Run Terminal Help ALSU_SHIFT Generate Simulate
ALSU_coverage.v > {} ALSU_coverage_pkg
1 package ALSU_coverage_pkg;
7 class ALSU_coverage extends uvm_component;
105 //5
106 walkingones:cross CB3,CB2,CB1{
107     option.cross_auto_bin_max=0;
108     bins arithA=binsof(CB3.Bins_bitwise) && binsof(CB2.B_data_0) && binsof(CB1.A_data_walkingones) ;
109     bins arithB=binsof(CB3.Bins_bitwise) && binsof(CB1.A_data_0) && binsof(CB2.B_data_walkingones) ;
110 }
111 invalidation:cross red_op_A,red_op_B,special{
112     option.cross_auto_bin_max=0;
113     bins RopA_notXor=binsof(special) intersect([ADD:ROTATE]) && binsof(red_op_A.red_op_A1) ;
114     bins RopB_notXor=binsof(special) intersect([ADD:ROTATE]) && binsof(red_op_B.red_op_B1) ;
115 }
116 endgroup
117
118 function new(string name ="coverage",uvm_component parent=null);
119     super.new(name,parent);
120     cvr_gp=new();
121 endfunction
122
123
124 function void build_phase(uvm_phase phase);
125     super.build_phase(phase);
126     cov_fifo=new("fifo",this);
127     cov_gp=new("cov_export",this);
128 endfunction
129
130 function void connect_phase(uvm_phase phase);
131     super.connect_phase(phase);
132     cov_gp.connect(cov_fifo.analysis_export);
133 endfunction
134
135 task run_phase(uvm_phase phase);
136     super.run_phase(phase);
137     forever begin
138         cov_fifo.get(item);
139         cvr_gp.sample();
    ...
endtask
endclass
endpackage
```



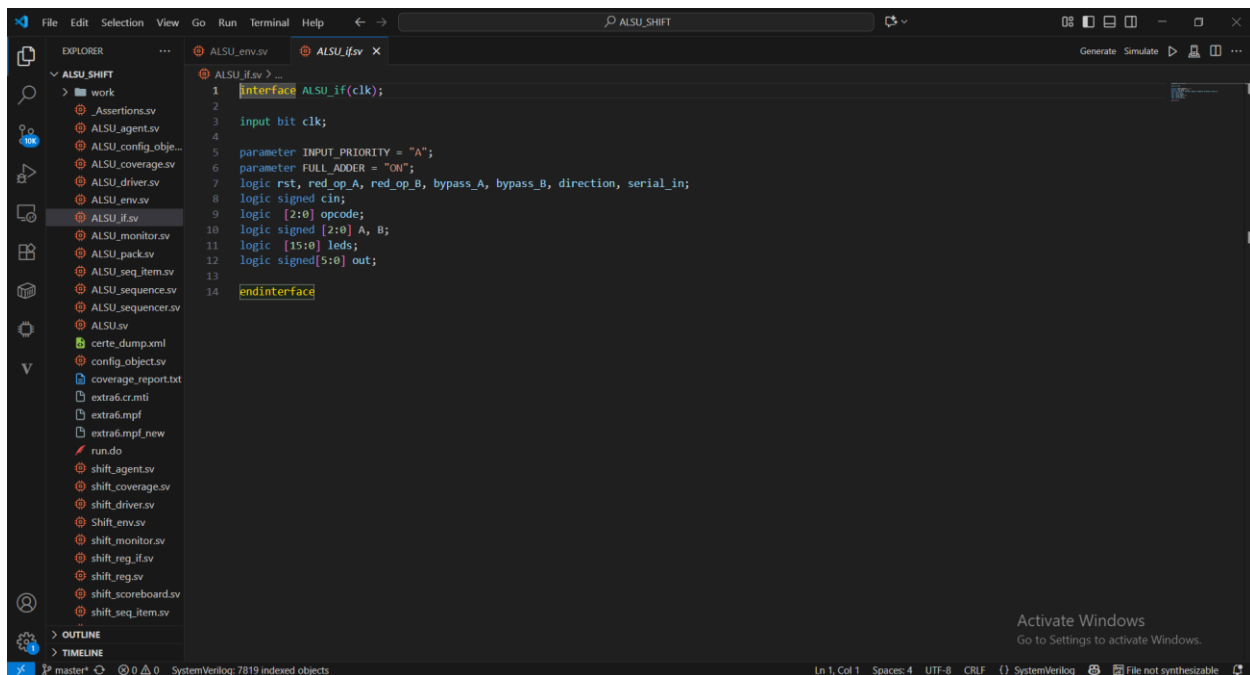
6. ALSU_driver



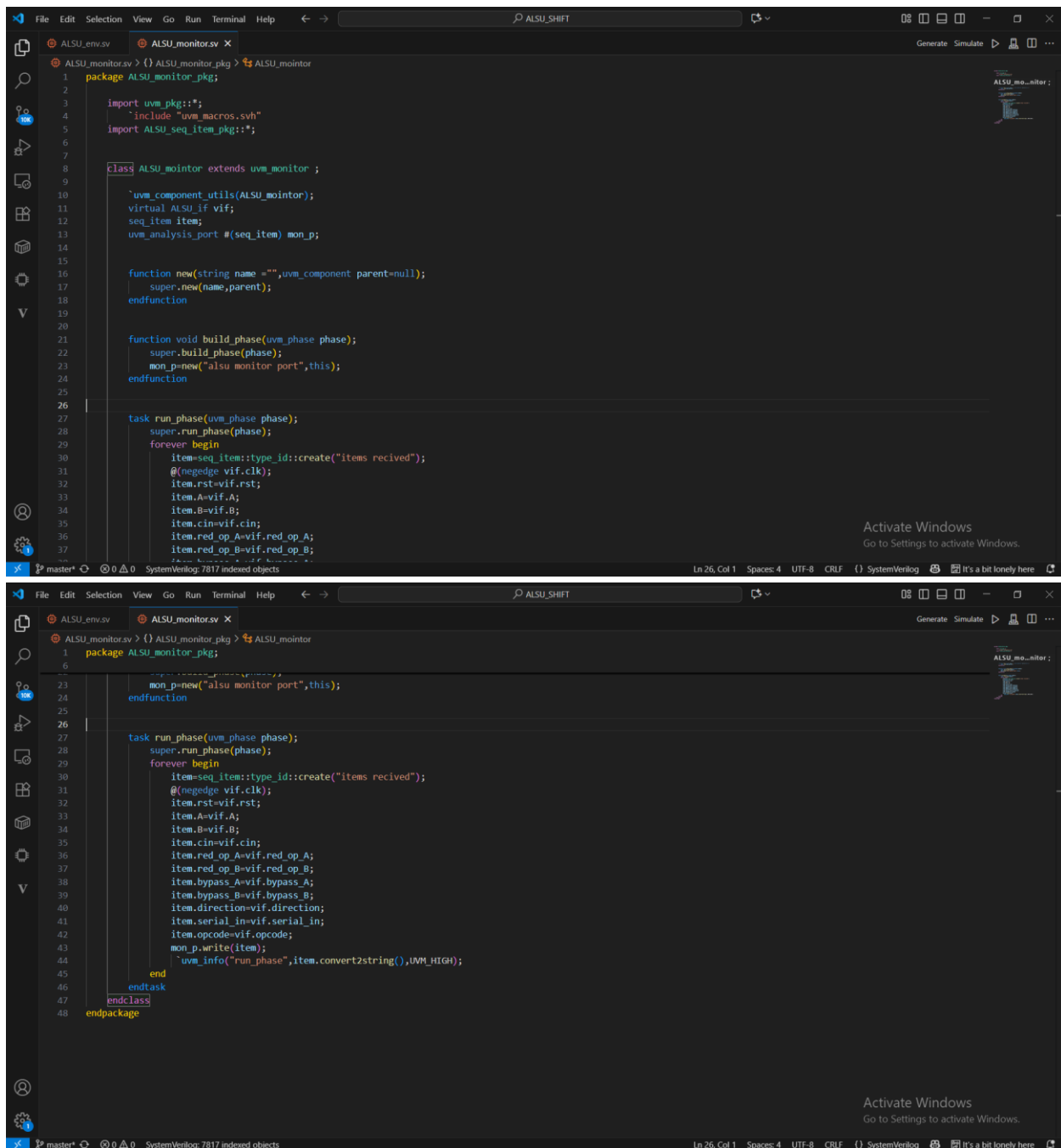
7. ALSU_env



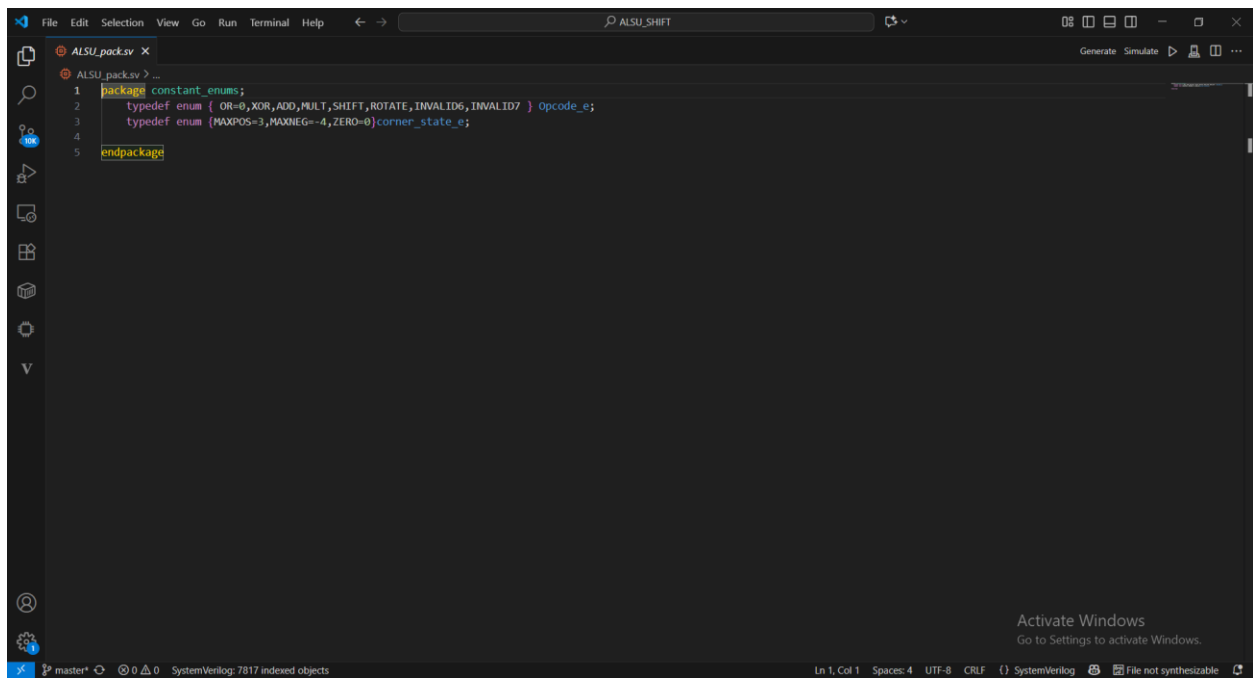
8. ALSU_if



9. ALSU_monitor



10. ALSU_shared_package



11. ALSU_seq_item

```

File Edit Selection View Go Run Terminal Help
ALSU_SHIFT
ALSU_seq_item.sv
1 package ALSU_seq_item_pkg;
2 import uvm_pkg::*;
3 `include "uvm_macros.svh"
4 import constant_enums::*;
5
6 class seq_item extends uvm_sequence_item;
7     `uvm_object_utils(seq_item);
8     rand bit clk, rst, red_op_A, red_op_B, bypass_A, bypass_B, direction, serial_in;
9     rand logic cin;
10    rand logic [2:0] opcode;
11    rand bit signed [2:0] A, B;
12    bit signed [5:0] out;
13    bit [15:0] leds;
14
15    bit [2:0] ones_number={3'b001,3'b010,3'b100};
16    rand bit [2:0] found,notfound;
17    rand corner_state e a_state;
18    rand bit [2:0] rem_numbers;
19    rand opcode e arr[6];
20    function new(string name = "seq_item");
21        super.new(name);
22    endfunction
23
24    function string convert2string();
25        return $formatf("%s reset =%0b cin=%0b ,red_op_A=%0b ,red_op_B=%0b ,bypass_A=%0b ,bypass_B=%0b ,
26            direction=%0b ,serial_in=%0b ,opcode=%0b ,A=%0b,B=%0b ,out =%0b,leds=%0b ",
27            super.convert2string(),rst,cin,red_op_A,red_op_B,bypass_A,bypass_B,direction,serial_in,opcode,A,B,out,leds);
28    endfunction
29
30    function string convert2string_stimulus();
31        return $formatf("%s reset =%0b cin=%0b ,red_op_A=%0b ,red_op_B=%0b ,bypass_A=%0b ,bypass_B=%0b ,
32            direction=%0b ,serial_in=%0b ,opcode=%0b ,A=%0b,B=%0b ",
33            super.convert2string(),rst,cin,red_op_A,red_op_B,bypass_A,bypass_B,direction,serial_in,opcode,A,B);
34    endfunction
35
36    constraint x {
37        rem_numbers!= MAXPOS||MAXNEG||ZERO;

```

```

File Edit Selection View Go Run Terminal Help
ALSU_SHIFT
ALSU_seq_item.sv
6 class seq_item extends uvm_sequence_item;
36    constraint x {
37        rem_numbers!= MAXPOS||MAXNEG||ZERO;
38        rst dist {1:=5 , 0:=95};
39
40        found inside {ones_number};
41        !(!notfound inside {ones_number});
42
43        if (opcode ==ADD || opcode== MULT){
44            A dist {a_state:=80,rem_numbers:=20};
45            B dist {a_state:=80,rem_numbers:=20};
46        }
47        if (opcode ==OR || opcode== XOR ){
48            if(red_op_A){
49                A dist {found:=80,notfound:=20};
50                B==3'b000;
51            }
52            else if (red_op_B){
53                B dist {found:=80,notfound:=20};
54                A==3'b000;
55            }
56        }
57
58        opcode dist {[OR:ROTATE]:=80,[INVALID6:INVALID7]};
59
60        bypass_A dist {0:=90,1:=10};
61        bypass_B dist {0:=90,1:=10};
62    }
63    constraint y{
64        unique(arr);
65        foreach(arr[i])
66            arr[i] inside {[OR:ROTATE]};
67    }
68 endclass
69 endpackage
70

```

12. ALSU_sequence

```
File Edit Selection View Go Run Terminal Help ALSU_SHIFT
ALSU_seq_item.v ALSU_sequence.v x
ALSU_sequence.v > ...
1 package ALSU_sequence_pkg;
2 import uvm_pkg::*;
3 `include "uvm_macros.svh"
4 import ALSU_seq_item_pkg::*;
5
6 class reset_sequence extends uvm_sequence #(seq_item);
7     `uvm_object_utils(reset_sequence);
8
9     function new(string name = "reset_sequence");
10         super.new(name);
11     endfunction
12
13     task body();
14         seq_item item;
15         item = seq_item::type_id::create("sequence reset");
16         start_item(item);
17         item.rst = 1;
18         finish_item(item);
19     endtask
20 endclass //ALSU_if
21
22 class main_sequence extends uvm_sequence #(seq_item);
23     `uvm_object_utils(main_sequence);
24
25     function new(string name = "main_sequence", uvm_component parent = null);
26         super.new(name);
27     endfunction
28
29     task body();
30
31         repeat(20000) begin
32             seq_item item;
33             item = seq_item::type_id::create("sequence");
34             item.constraint_mode(0);
35             item.x.constraint_mode(1);
36             start_item(item);
37             assert(item.randomize());
38         end
39     endtask
40 endclass
```

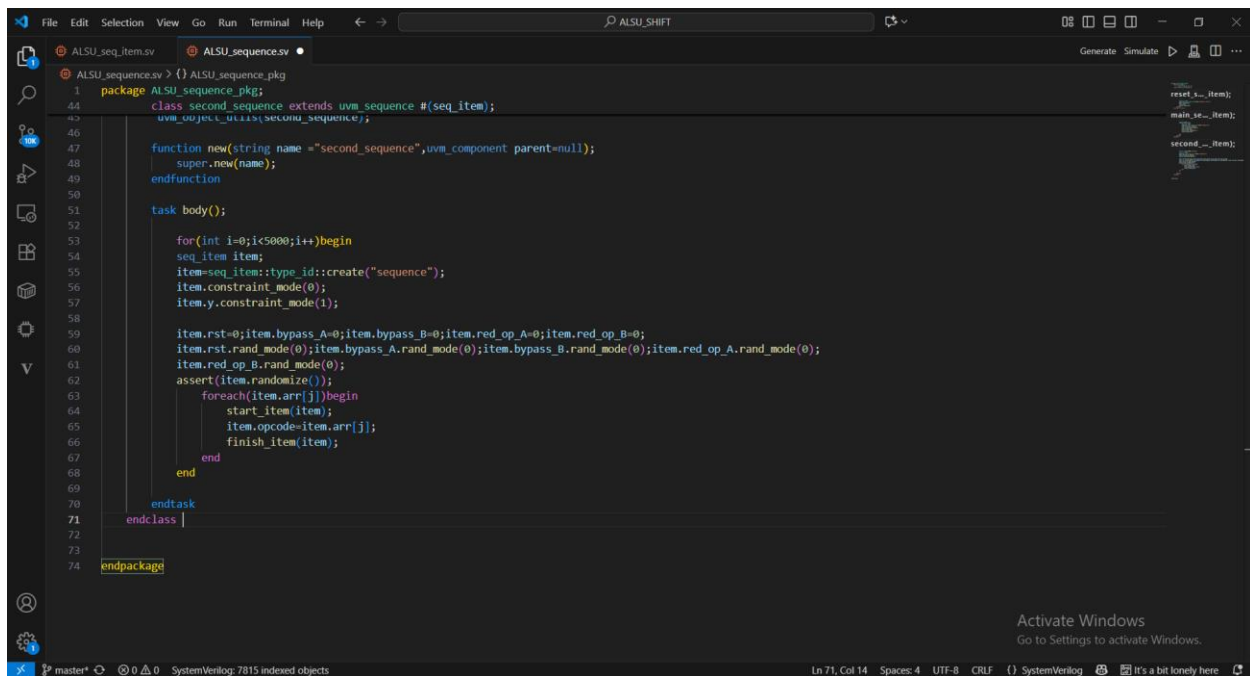
Activate Windows
Go to Settings to activate Windows.

Ln 1, Col 1 Spaces: 4 UTF-8 CRLF SystemVerilog File not synthesizable

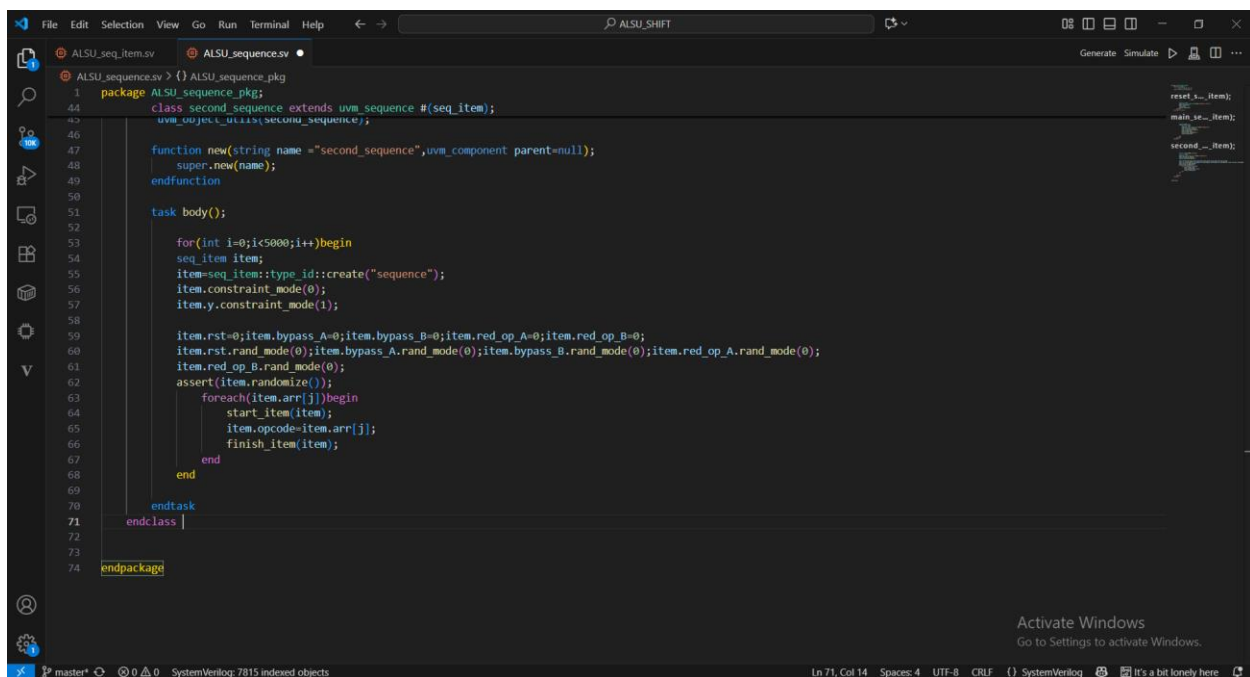
```
File Edit Selection View Go Run Terminal Help ALSU_SHIFT
ALSU_seq_item.v ALSU_sequence.v
ALSU_sequence.v > {} ALSU_sequence_pkg > main_sequence
1 package ALSU_sequence_pkg;
22 class main_sequence extends uvm_sequence #(seq_item);
28
29     task body();
30
31         repeat(20000) begin
32             seq_item item;
33             item = seq_item::type_id::create("sequence");
34             item.constraint_mode(0);
35             item.x.constraint_mode(1);
36             start_item(item);
37             assert(item.randomize());
38             finish_item(item);
39         end
40     endtask
41 endclass
42
43 class second_sequence extends uvm_sequence #(seq_item);
44     `uvm_object_utils(second_sequence);
45
46     function new(string name = "second_sequence", uvm_component parent = null);
47         super.new(name);
48     endfunction
49
50     task body();
51
52         for(int i=0; i<5000; i++) begin
53             seq_item item;
54             item = seq_item::type_id::create("sequence");
55             item.constraint_mode(0);
56             item.y.constraint_mode(1);
57
58             item.rst = 0; item.bypass_A = 0; item.bypass_B = 0; item.red_op_A = 0; item.red_op_B = 0;
59             item.rst.rand_mode(0); item.bypass_A.rand_mode(0); item.bypass_B.rand_mode(0); item.red_op_A.rand_mode(0);
60             item.red_op_B.rand_mode(0);
61             assert(item.randomize());
62         end
63     endtask
64 endclass
```

Activate Windows
Go to Settings to activate Windows.

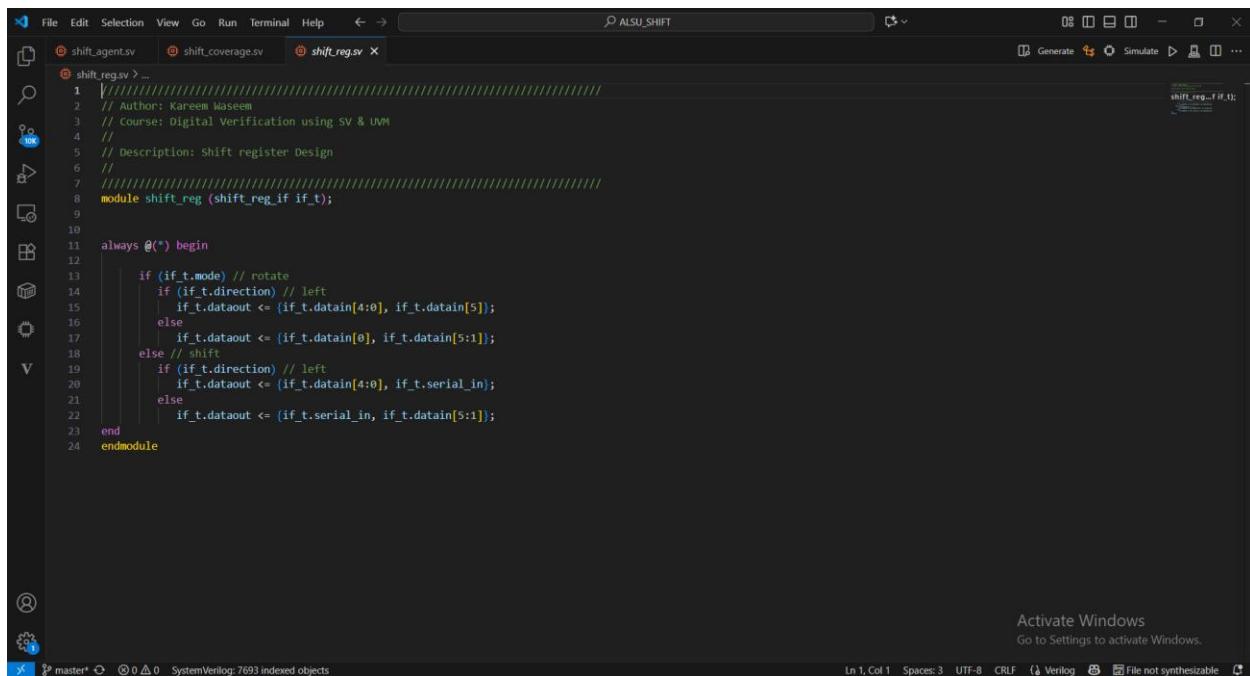
Ln 29, Col 21 Spaces: 4 UTF-8 CRLF SystemVerilog It's a bit lonely here



13. ALSU_sequencer



14. shift_reg_design

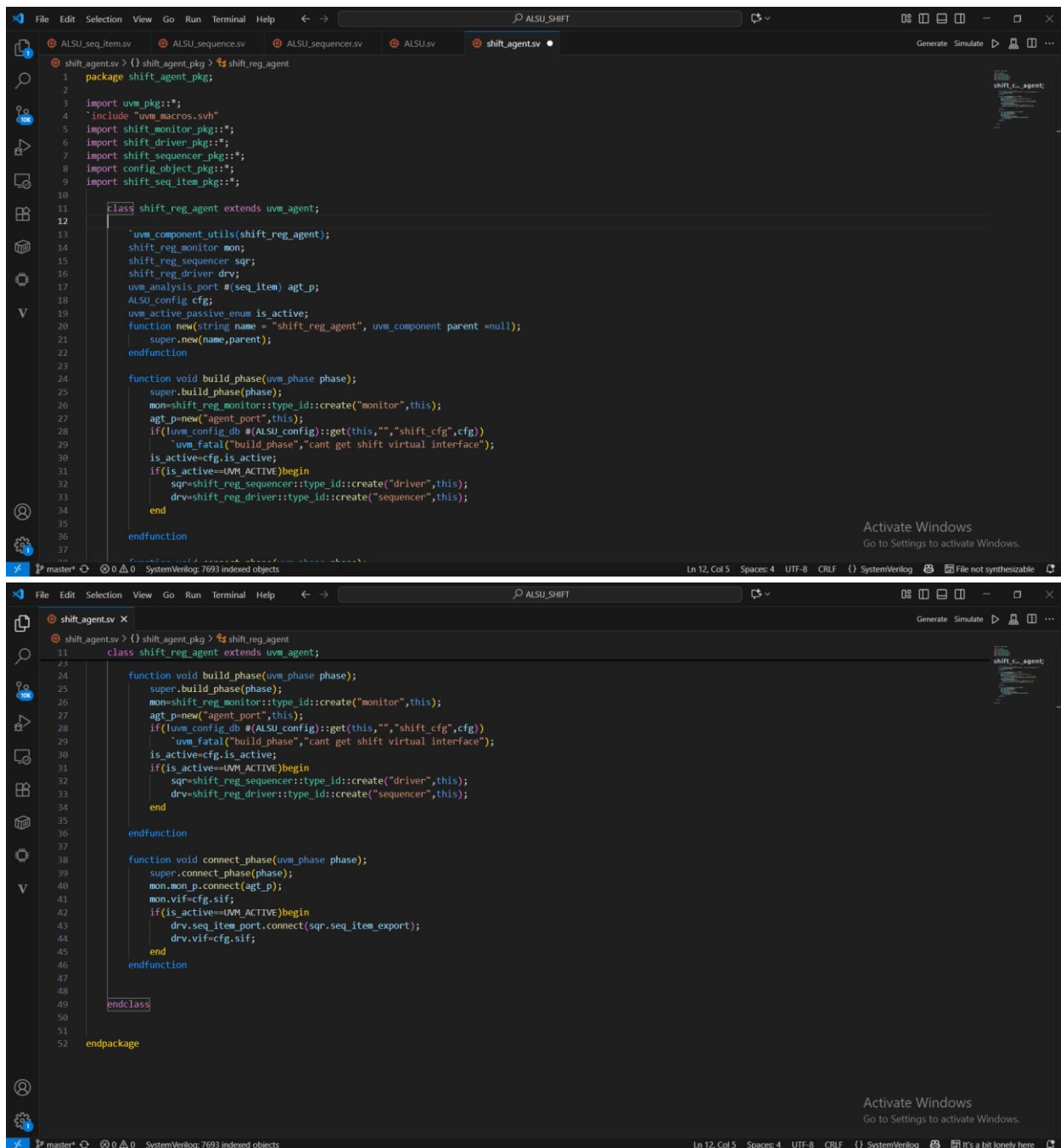


The screenshot shows a code editor window with the title bar "ALSU_SHIFT". The editor has three tabs: "shift_reg_agent.sv", "shift_coverage.sv", and "shift_reg.sv". The "shift_reg.sv" tab is active, displaying the following SystemVerilog code:

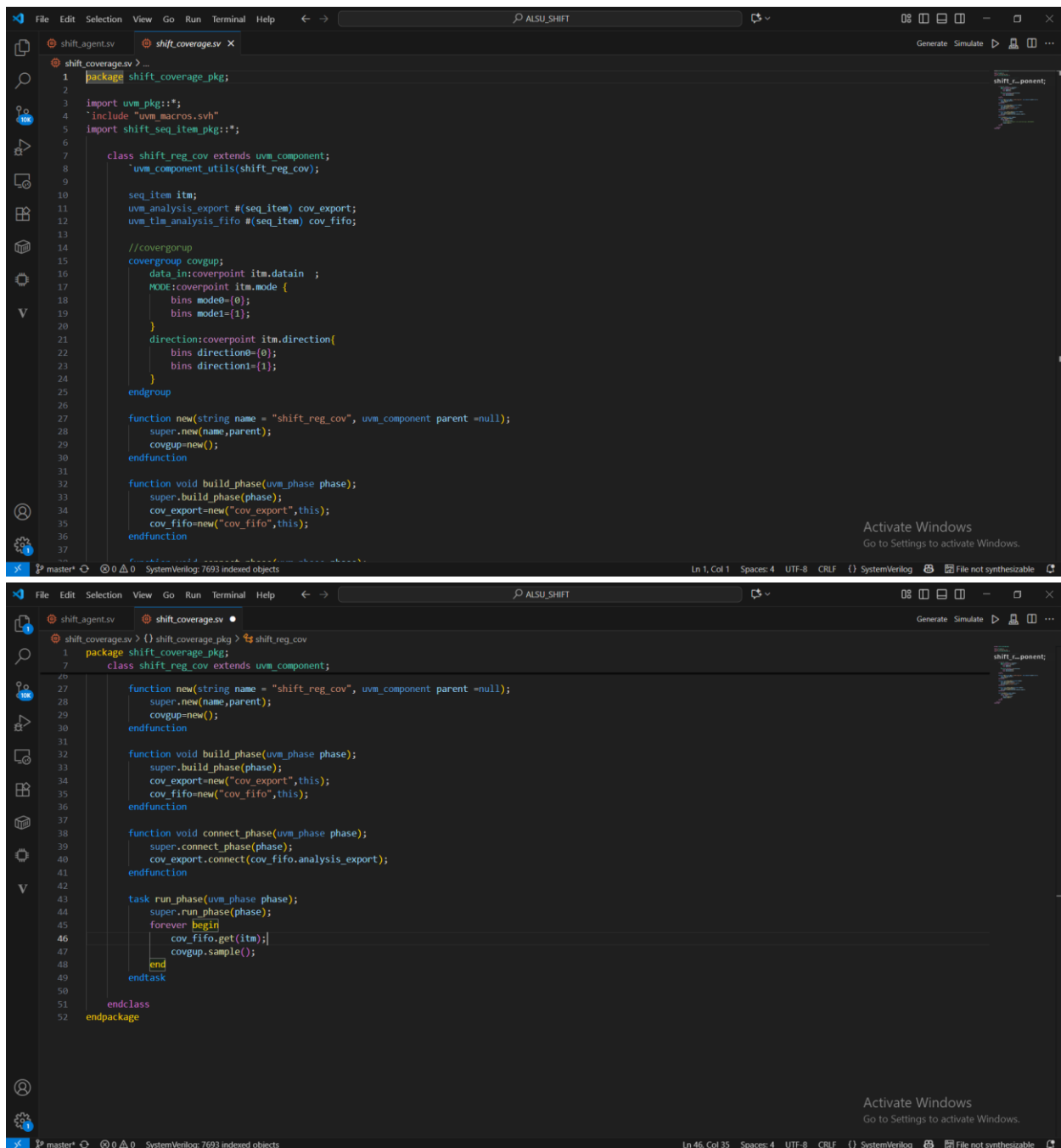
```
1 //////////////////////////////////////////////////////////////////////////////////////////////////////////////////
2 // Author: Kareem Waseem
3 // Course: Digital Verification using SV & UVM
4 //
5 // Description: Shift register Design
6 //
7 //////////////////////////////////////////////////////////////////////////////////////////////////////////////////
8 module shift_reg (shift_reg_if if_t);
9
10
11 always @(*) begin
12
13     if (if_t.mode) // rotate
14         if (if_t.direction) // left
15             if_t.dataout <= {if_t.datain[4:0], if_t.datain[5]};
16         else
17             if_t.dataout <= {if_t.datain[0], if_t.datain[5:1]};
18         else // shift
19             if (if_t.direction) // left
20                 if_t.dataout <= {if_t.datain[4:0], if_t.serial_in};
21             else
22                 if_t.dataout <= {if_t.serial_in, if_t.datain[5:1]};
23         end
24 endmodule
```

The status bar at the bottom indicates "Ln 1, Col 1", "Spaces: 3", "UTF-8", "CRLF", "Verilog", and "File not synthesizable".

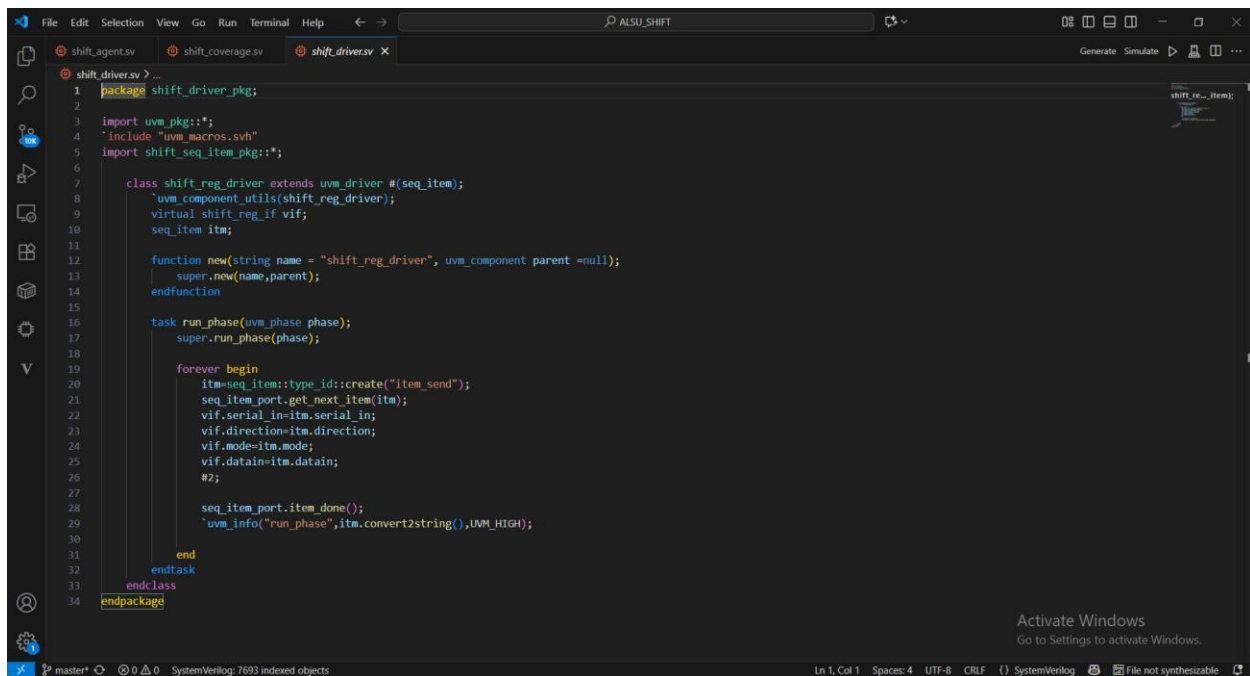
15. shift_reg_agent



16. shift_reg_coverage



17. shift_reg_driver

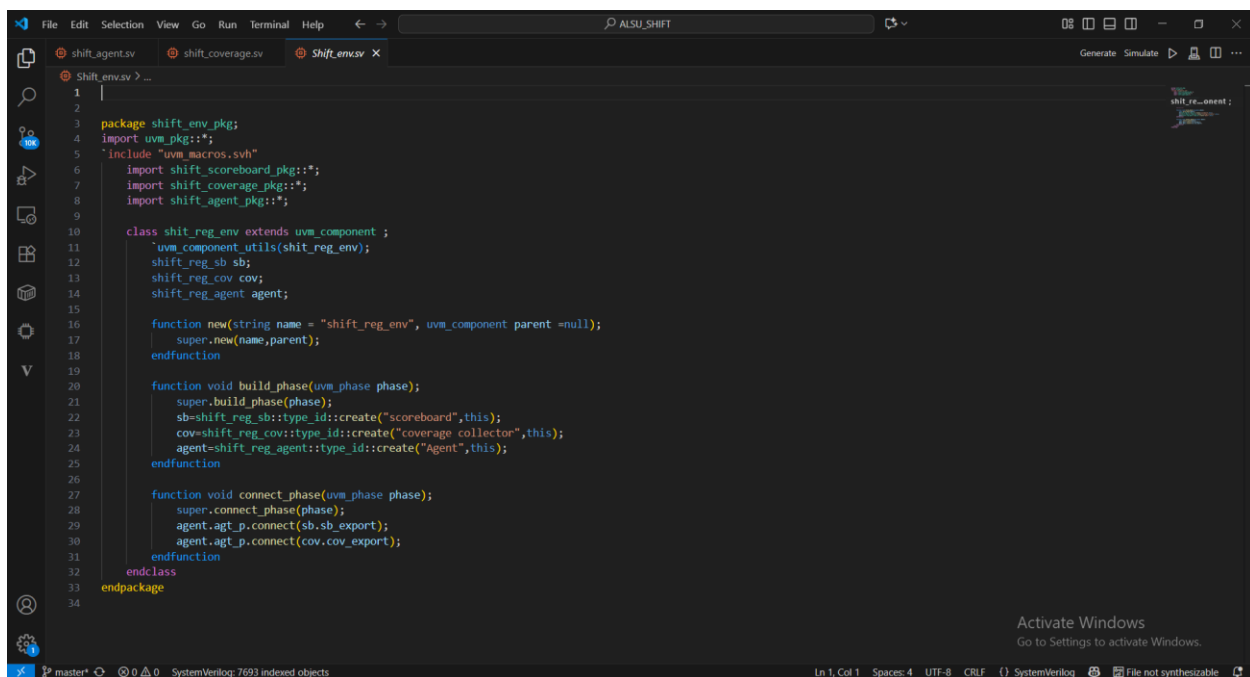


```
1 package shift_driver_pkg;
2
3 import uvm_pkg::*;
4 `include "uvm_macros.svh"
5 import shift_seq_item_pkg::*;
6
7 class shift_reg_driver extends uvm_driver #(seq_item);
8     uvm_component_utils(shift_reg_driver);
9     virtual shift_reg_if vif;
10    seq_item itm;
11
12    function new(string name = "shift_reg_driver", uvm_component parent = null);
13        super.new(name,parent);
14    endfunction
15
16    task run_phase(uvm_phase phase);
17        super.run_phase(phase);
18
19        forever begin
20            itm=seq_item::type_id::create("item_send");
21            seq_item_port.get_next_item(itm);
22            vif.serial_in=itm.serial_in;
23            vif.direction=itm.direction;
24            vif.mode=itm.mode;
25            vif.datain=itm.datain;
26            #2;
27
28            seq_item_port.item_done();
29            "uvm_info("run_phase",itm.convert2string(),UVM_HIGH);
30
31        end
32    endtask
33 endclass
34 endpackage
```

Activate Windows
Go to Settings to activate Windows.

master* 0 0 SystemVerilog: 7693 indexed objects Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} SystemVerilog File not synthesizable

18. Shift_env

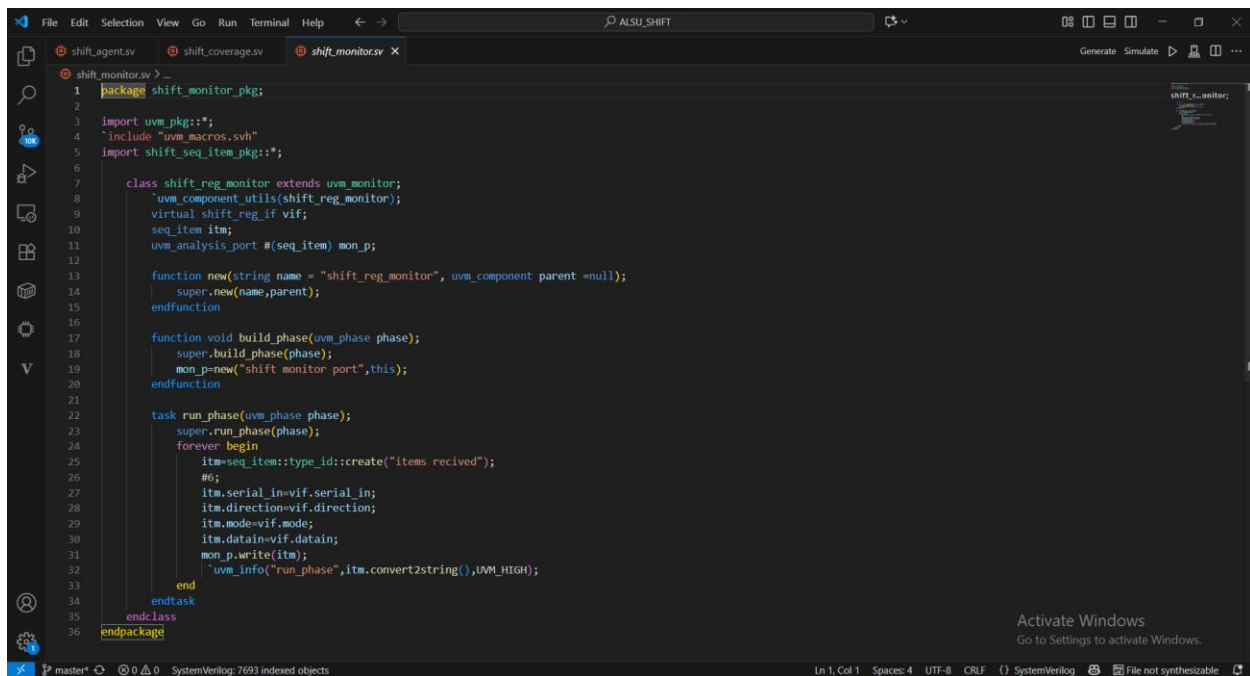


```
1 |
2
3 package shift_env_pkg;
4 import uvm_pkg::*;
5 `include "uvm_macros.svh"
6 import shift_scoreboard_pkg::*;
7 import shift_coverage_pkg::*;
8 import shift_agent_pkg::*;
9
10 class shift_reg_env extends uvm_component ;
11     uvm_component_utils(shift_reg_env);
12     shift_reg_sb sb;
13     shift_reg_cov cov;
14     shift_reg_agent agent;
15
16     function new(string name = "shift_reg_env", uvm_component parent = null);
17         super.new(name,parent);
18     endfunction
19
20     function void build_phase(uvm_phase phase);
21         super.build_phase(phase);
22         sb=shift_reg_sb::type_id::create("scoreboard",this);
23         cov=shift_reg_cov::type_id::create("coverage collector",this);
24         agent=shift_reg_agent::type_id::create("Agent",this);
25     endfunction
26
27     function void connect_phase(uvm_phase phase);
28         super.connect_phase(phase);
29         agent.agt_p.connect(sb.sb_export);
30         agent.agt_p.connect(cov.cov_export);
31     endfunction
32 endclass
33 endpackage
34
```

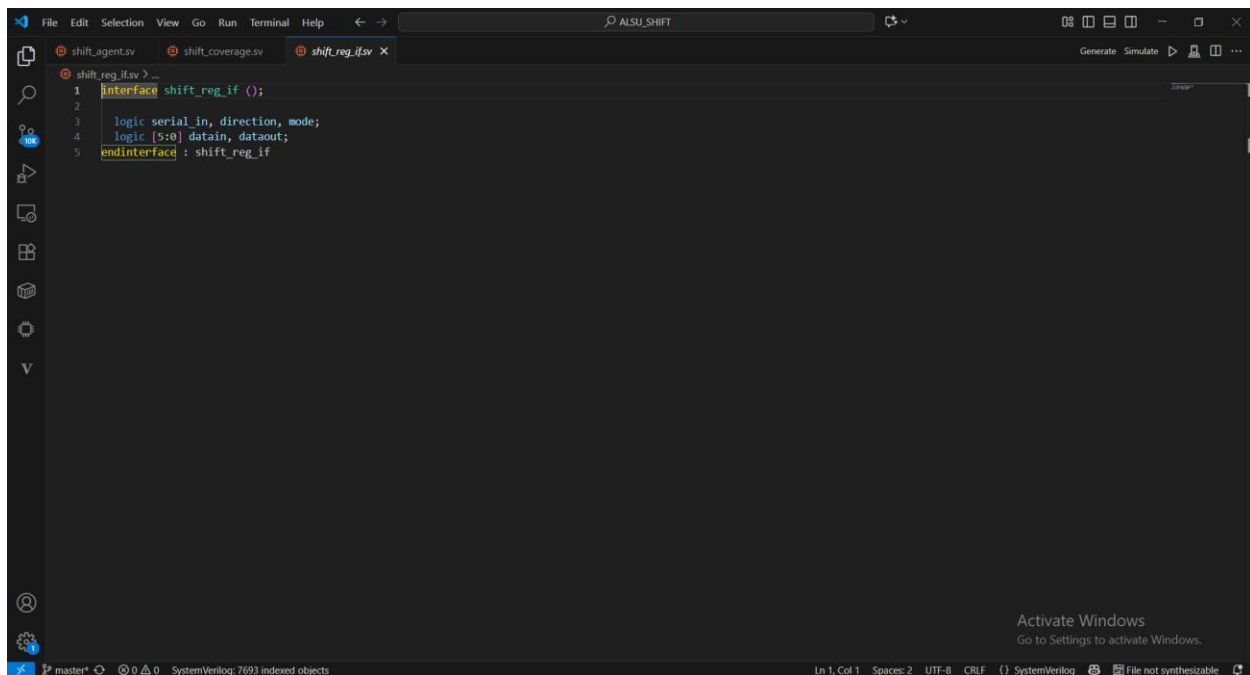
Activate Windows
Go to Settings to activate Windows.

master* 0 0 SystemVerilog: 7693 indexed objects Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} SystemVerilog File not synthesizable

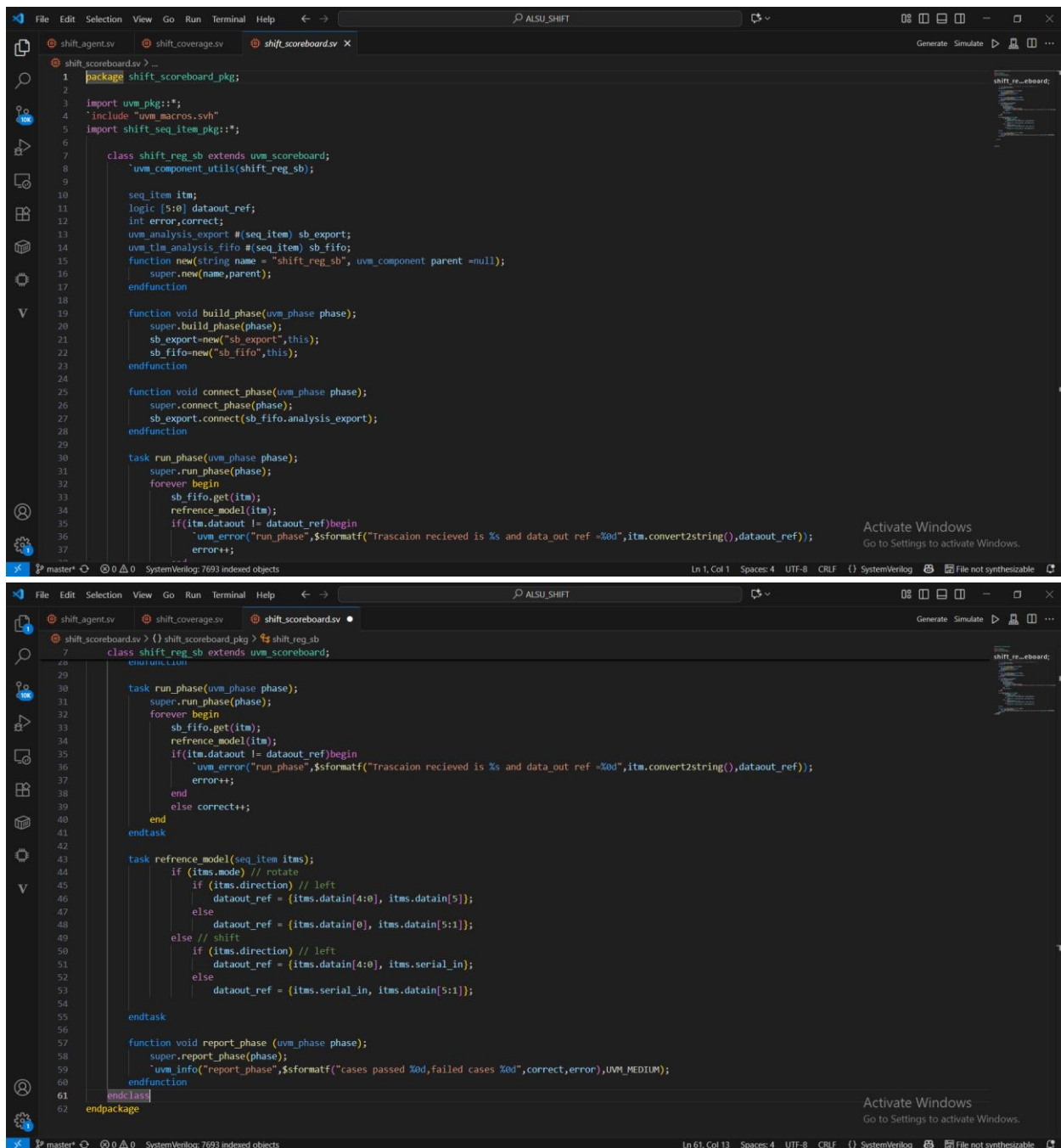
19. shift_monitor



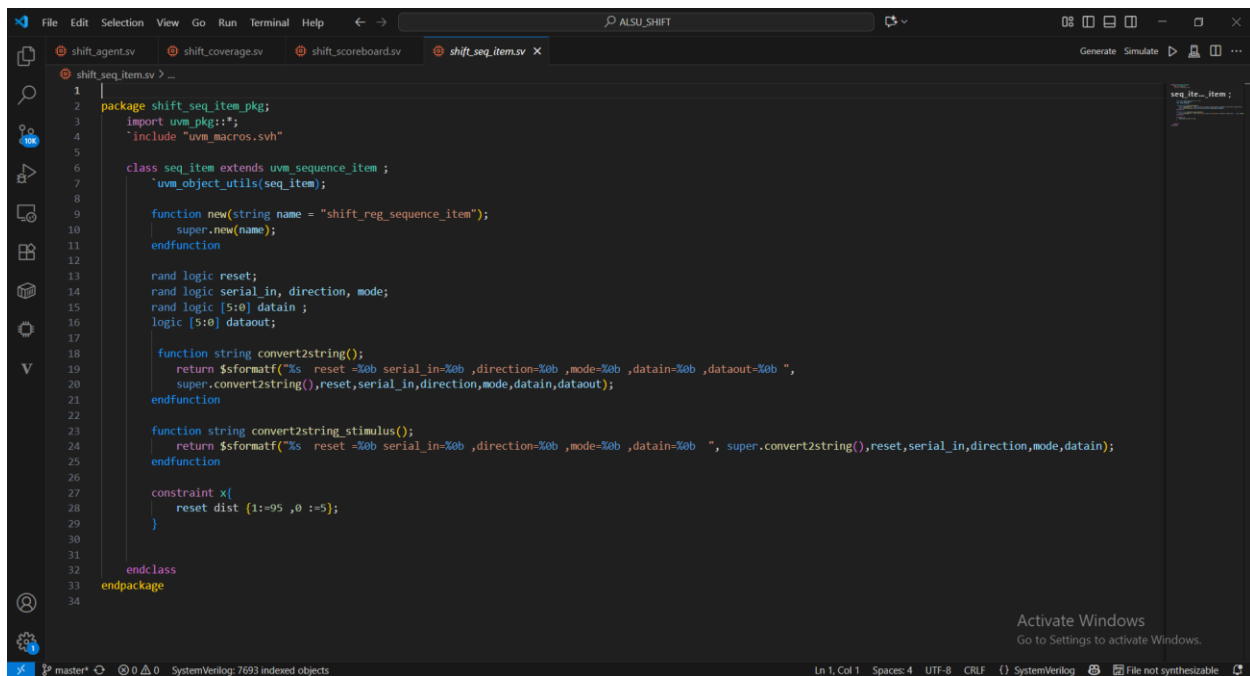
20. shift_reg_if



21. shift_scoreboard



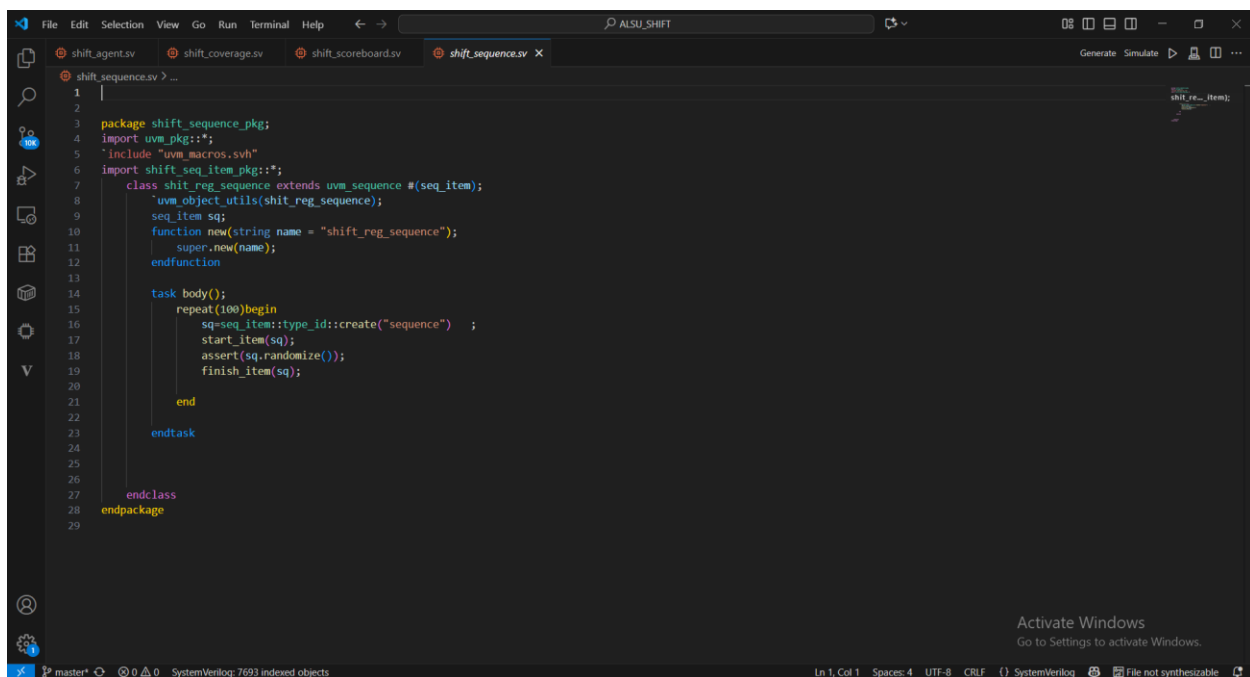
22. shift_seq_item



The screenshot shows the VS Code editor with the ALSU_SHIFT project open. The file explorer on the left lists several files: shift_agent.sv, shift_coverage.sv, shift_scoreboard.sv, and shift_seq_item.sv. The main editor window displays the contents of shift_seq_item.sv, which defines a Verilog class for sequence items. The code includes package declarations, imports, and a class definition with various methods and constraints.

```
1 |
2 | package shift_seq_item_pkg;
3 | import uvm_pkg::*;
4 | `include "uvm_macros.svh"
5 |
6 | class seq_item extends uvm_sequence_item ;
7 |     `uvm_object_utils(seq_item);
8 |
9 |     function new(string name = "shift_reg_sequence_item");
10 |         super.new(name);
11 |     endfunction
12 |
13 |     rand logic reset;
14 |     rand logic serial_in, direction, mode;
15 |     rand logic [5:0] datain ;
16 |     logic [5:0] dataout;
17 |
18 |     function string convert2string();
19 |         return $formatf("%s reset=%0b serial_in=%0b ,direction=%0b ,mode=%0b ,datain=%0b ,dataout=%0b ",
20 |             super.convert2string(),reset,serial_in,direction,mode,datain,dataout);
21 |     endfunction
22 |
23 |     function string convert2string_stimulus();
24 |         return $formatf("%s reset=%0b serial_in=%0b ,direction=%0b ,mode=%0b ,datain=%0b ", super.convert2string(),reset,serial_in,direction,mode,datain);
25 |     endfunction
26 |
27 |     constraint x{
28 |         reset dist {1:=95 ,0 :=5};
29 |     }
30 |
31 | endclass
32 | endpackage
33 |
34 |
```

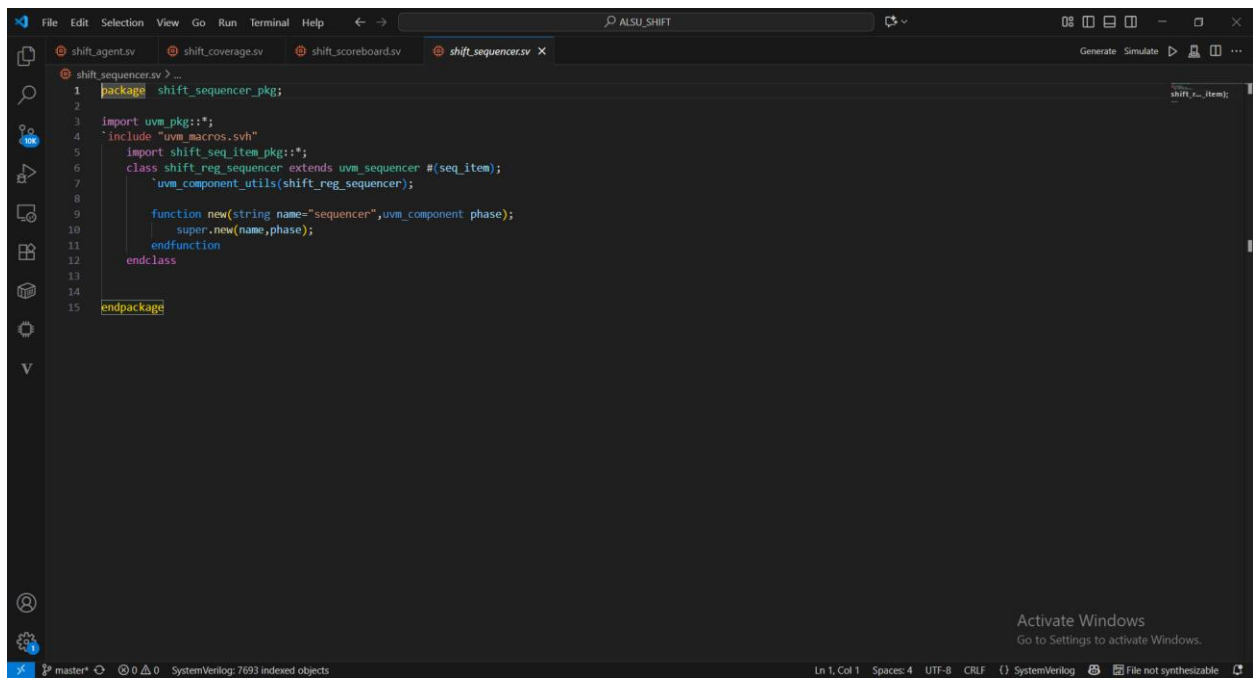
23. shift_sequence



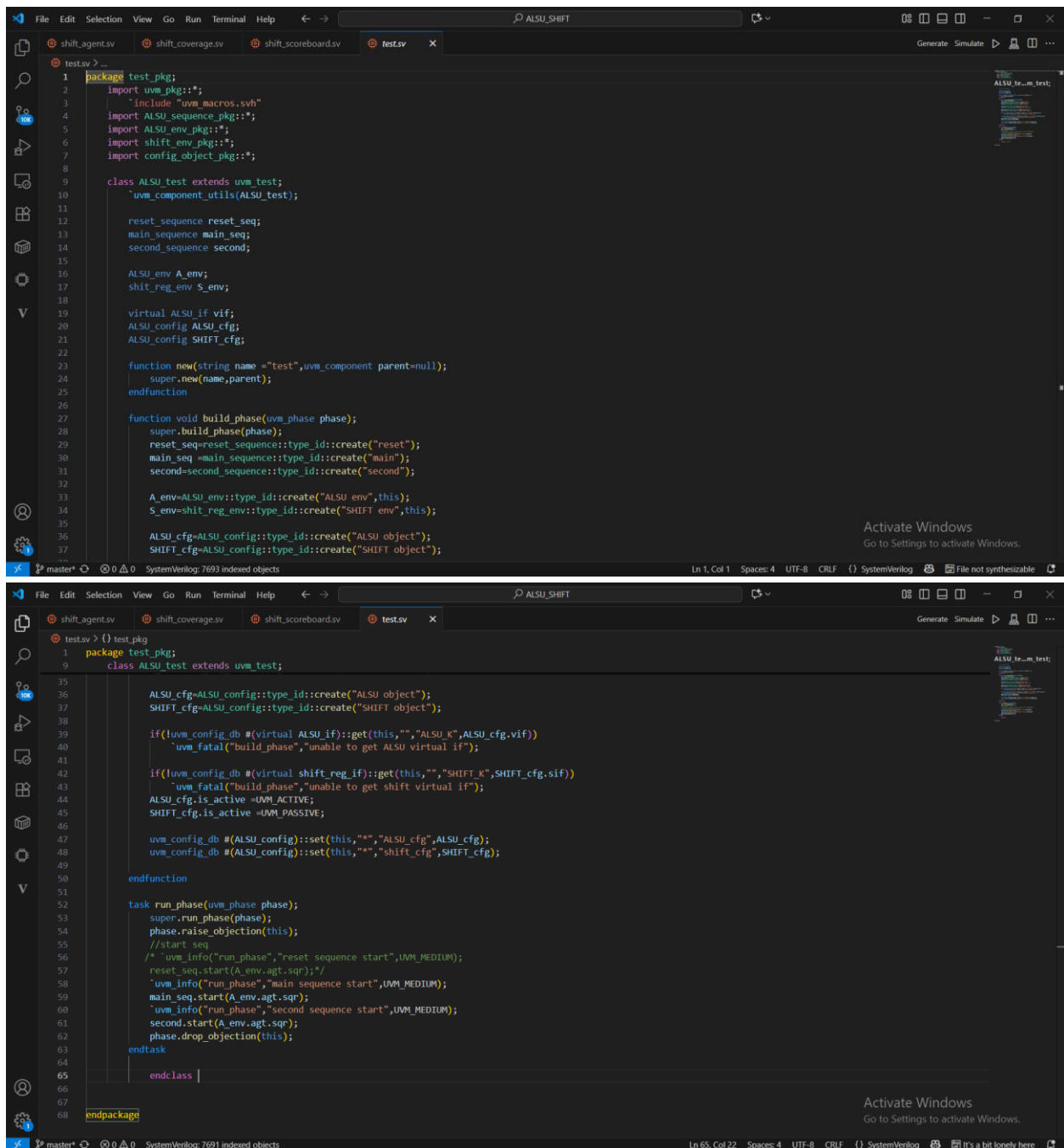
The screenshot shows the VS Code editor with the ALSU_SHIFT project open. The file explorer on the left lists several files: shift_agent.sv, shift_coverage.sv, shift_scoreboard.sv, and shift_sequence.sv. The main editor window displays the contents of shift_sequence.sv, which defines a Verilog class for a sequence of shift register operations. The code includes package declarations, imports, and a class definition with a task method for generating the sequence.

```
1 |
2 | package shift_sequence_pkg;
3 | import uvm_pkg::*;
4 | `include "uvm_macros.svh"
5 | import shift_seq_item_pkg::*;
6 |
7 | class shift_reg_sequence extends uvm_sequence #(seq_item);
8 |     `uvm_object_utils(shift_reg_sequence);
9 |     seq_item sq;
10 |     function new(string name = "shift_reg_sequence");
11 |         super.new(name);
12 |     endfunction
13 |
14 |     task body();
15 |         repeat(100)begin
16 |             sq=seq_item::type_id::create("sequence") ;
17 |             start_item(sq);
18 |             assert(sq.randomize());
19 |             finish_item(sq);
20 |         end
21 |     endtask
22 |
23 | endclass
24 | endpackage
25 |
26 |
27 |
28 |
29 |
```

24. shift_sequencer



25. test



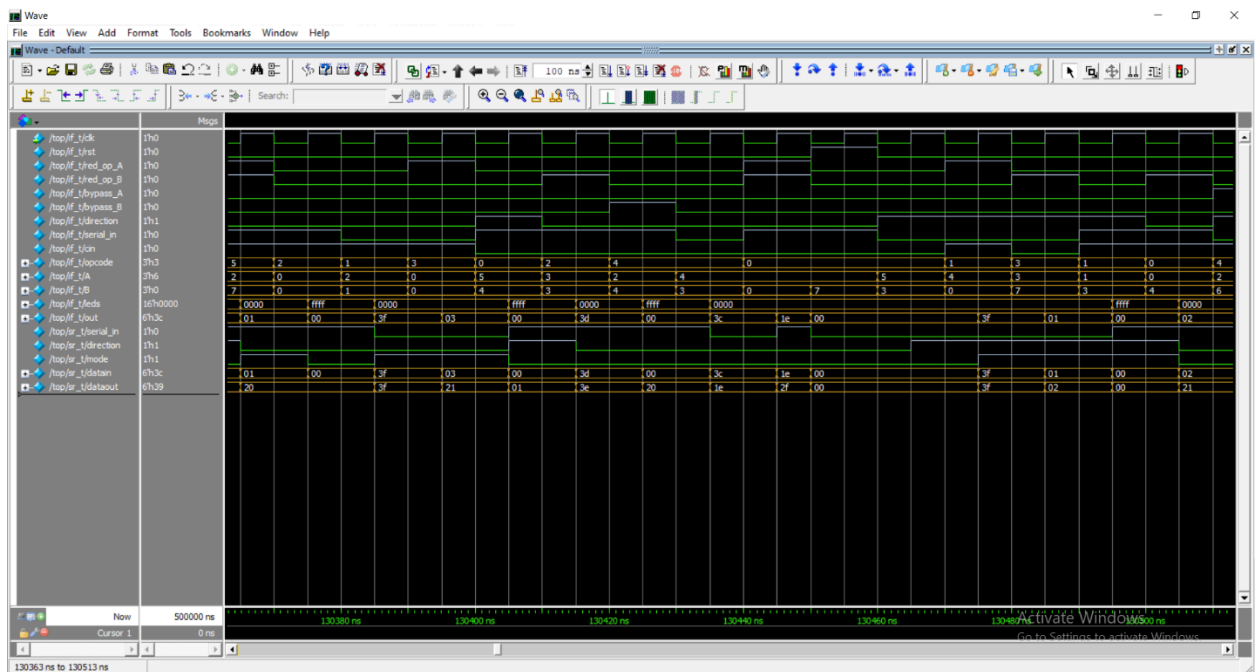
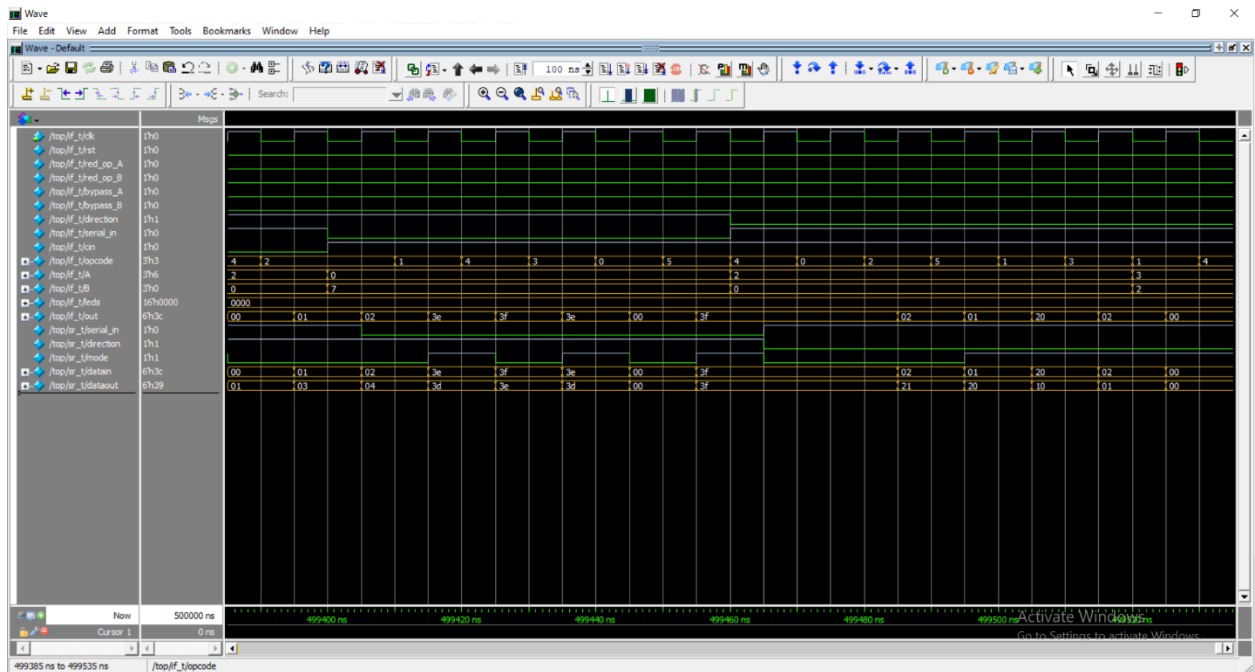
26. top

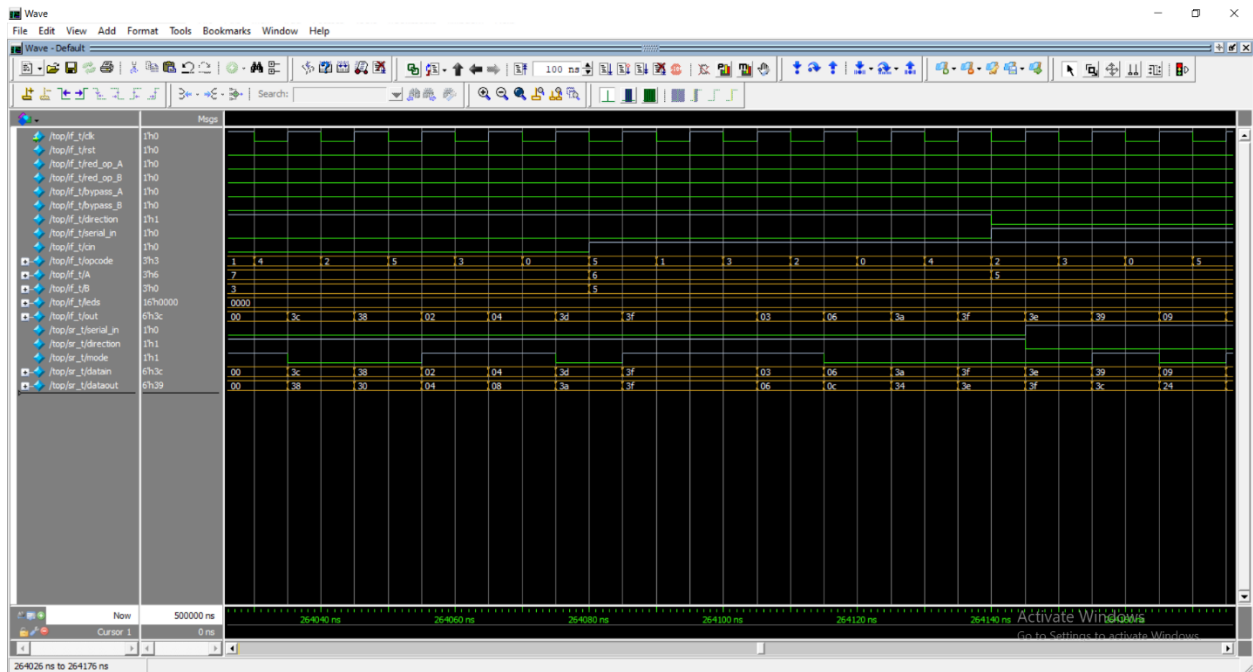
```
File Edit Selection View Go Run Terminal Help
ALSU_SHIFT
shift_agent.v shift_coverage.v shift_scoreboard.v test.v top.v
Generate Simulate
top.v
1 import uvm_pkg::*;
2 `include "uvm_macros.svh"
3 import test_pkg::*;
4
5 module top();
6 bit clk=0;
7 always #5 clk=1clk;
8
9 ALSU_if if_t(clk);
10 shift_reg_if sr_t();
11 shift_reg sr(sr_t);
12 ALSU_DUT if_t, sr_t);
13
14 bind ALSU Assertions AS(if_t);
15
16 initial begin
17     uvm_config_db #(virtual ALSU_if)::set(null,"", "ALSU_K", if_t);
18     uvm_config_db #(virtual shift_reg_if)::set(null,"", "SHIFT_K", sr_t);
19     run_test("ALSU_test");
20 end
21
22 endmodule
23
24
SystemVerilog: 7691 indexed objects
Ln 24, Col 1 Spaces: 4 UTF-8 CRLF Verilog File not synthesizable
Activate Windows
Go to Settings to activate Windows.
```

Do file

```
File Edit Selection View Go Run Terminal Help
ALSU_SHIFT
ALSU_seq_item.v ALSU_sequence.v ALSU_sequencer.v ALSU.v run.do
Generate Simulate
run.do
1 vlib work
2 vlog *v +cover -covercells
3 vsim -voptargs=-racc work.top -classdebug -uvmcontrol=all -cover
4
5 add wave -position insertpoint sim:/top/if_t/*
6 add wave -position insertpoint sim:/top/sr_t/*
7 coverage save Top_tb.ucdb -onexit
8 run -all
9
10
SystemVerilog: 7693 indexed objects
Ln 10, Col 1 Spaces: 4 UTF-8 CRLF Plain Text It's a bit lonely here
Activate Windows
Go to Settings to activate Windows.
```

Waveform





Reports

assert

coverage_report - Notepad				
File Edit Format View Help				
Assertion Coverage: 24 20 4 83.33%				
Assertions				
Name	File(Line)	Failure Count	Pass Count	
/top/DUT/AS/rst_check	Assertions.sv(9)	0	1	
/top/DUT/AS/leds_check	Assertions.sv(185)	0	1	
/top/DUT/AS/bypass1_check	Assertions.sv(187)	0	1	
/top/DUT/AS/bypass2_check	Assertions.sv(188)	0	0	
/top/DUT/AS/bypass3_check	Assertions.sv(189)	0	1	
/top/DUT/AS/bypass4_check	Assertions.sv(190)	0	1	
/top/DUT/AS/INVALID_check	Assertions.sv(193)	0	1	
/top/DUT/AS/OR1_check	Assertions.sv(199)	0	1	
/top/DUT/AS/OR2_check	Assertions.sv(200)	0	0	
/top/DUT/AS/OR3_check	Assertions.sv(201)	0	1	
/top/DUT/AS/OR4_check	Assertions.sv(202)	0	1	
/top/DUT/AS/OR5_check	Assertions.sv(203)	0	1	
/top/DUT/AS/XOR1_check	Assertions.sv(207)	0	1	
/top/DUT/AS/XOR2_check	Assertions.sv(208)	0	0	
/top/DUT/AS/XOR3_check	Assertions.sv(209)	0	1	
/top/DUT/AS/XOR4_check	Assertions.sv(210)	0	1	
/top/DUT/AS/XOR5_check	Assertions.sv(211)	0	1	
/top/DUT/AS/FullADD_check	Assertions.sv(215)	0	1	
Ln 43, Col 17 100% Windows (CRLF) UTF-8				
coverage_report - Notepad				
File Edit Format View Help				
/top/DUT/AS/XOR1_check	Assertions.sv(203)	0	1	
/top/DUT/AS/XOR2_check	Assertions.sv(207)	0	1	
/top/DUT/AS/XOR3_check	Assertions.sv(208)	0	0	
/top/DUT/AS/XOR4_check	Assertions.sv(209)	0	1	
/top/DUT/AS/XOR5_check	Assertions.sv(210)	0	1	
/top/DUT/AS/FullADD_check	Assertions.sv(211)	0	1	
/top/DUT/AS/FullADD_check	Assertions.sv(215)	0	1	
/top/DUT/AS/halfADD_check	Assertions.sv(216)	0	0	
/top/DUT/AS/MULT_check	Assertions.sv(220)	0	1	
/top/DUT/AS/shift_R_check	Assertions.sv(224)	0	1	
/top/DUT/AS/shift_L_check	Assertions.sv(225)	0	1	
/top/DUT/AS/rotate_R_check	Assertions.sv(229)	0	1	
/top/DUT/AS/rotate_L_check	Assertions.sv(230)	0	1	
Branch Coverage:				
Enabled Coverage	Bins	Hits	Misses	Coverage
Branches	2	2	0	100.00%
=====Branch Details=====				
Branch Coverage for instance /top/DUT/AS				
Line	Item	Count	Source	
File	_Assertions.sv			
8	1	44680	Count coming in to IF	
8		1648		
		42952	All False Count	
Branch totals: 2 hits of 2 branches = 100.00%				
Ln 160, Col 51 100% Windows (CRLF) UTF-8				

coverage_report - Notepad

File Edit Format View Help

Expression Coverage:
Enabled Coverage

Expressions 5 5 0 100.00%

=====Expression Details=====

Expression Coverage for instance /top/DUT/AS --

File _Assertions.sv

-----Focused Expression View-----
Line 5 Item 1 (((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) || ((if_t.opcode > 1) && (if_t.red_op_A | if_t.red_op_B)))
Expression totals: 5 of 5 input terms covered = 100.00%

Input Term Covered Reason for no coverage Hint

(if_t.opcode == INVALID6) Y
(if_t.opcode == INVALID7) Y
(if_t.opcode > 1) Y
if_t.red_op_A Y
if_t.red_op_B Y

Rows: Hits FEC Target Non-masking condition(s)

Row 1: 1 (if_t.opcode == INVALID6)_0 (~((if_t.opcode > 1) && (if_t.red_op_A | if_t.red_op_B)) && ~(if_t.opcode == INVALID7))
Row 2: 1 (if_t.opcode == INVALID6)_1 -
Row 3: 1 (if_t.opcode == INVALID7)_0 (~((if_t.opcode > 1) && (if_t.red_op_A | if_t.red_op_B)) && ~(if_t.opcode == INVALID6))
Row 4: 1 (if_t.opcode == INVALID7)_1 ~(if_t.opcode == INVALID6)
Row 5: 1 (if_t.opcode > 1)_0 ~(if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)
Row 6: 1 (if_t.opcode > 1)_1 (~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) && (if_t.red_op_A | if_t.red_op_B))
Row 7: 1 if_t.red_op_A_0 (~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) && (if_t.opcode > 1) && ~if_t.red_op_B)
Row 8: 1 if_t.red_op_A_1 (~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) && (if_t.opcode > 1) && ~if_t.red_op_B)
Row 9: 1 if_t.red_op_B_0 (~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) && (if_t.opcode > 1) && ~if_t.red_op_A)
Row 10: 1 if_t.red_op_B_1 (~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) && (if_t.opcode > 1) && ~if_t.red_op_A)

Statement Coverage:
Enabled Coverage

Statements 2 2 0 100.00%

=====Statement Details=====

coverage_report - Notepad

File Edit Format View Help

Row 10: 1 if_t.red_op_B_1 (~((if_t.opcode == INVALID6) || (if_t.opcode == INVALID7)) && (if_t.opcode > 1) && ~if_t.red_op_A)

Statement Coverage:
Enabled Coverage

Statements 2 2 0 100.00%

=====Statement Details=====

Statement Coverage for instance /top/DUT/AS --

Line Item Count Source

File _Assertions.sv
5 1 48005
7 1 44600

Toggle Coverage:
Enabled Coverage

Toggles 2 2 0 100.00%

=====Toggle Details=====

Toggle Coverage for instance /top/DUT/AS --

Node 1H->0L 0L->1H "Coverage"

invalid 1 1 100.00

Total Node Count = 1
Toggled Node Count = 1
Untoggled Node Count = 0

Toggle Coverage = 100.00% (2 of 2 bins)

=====

Instance: /top/DUT
Design Unit: work.ALSU
=====

Branch Coverage:
Enabled Coverage

Bins Hits Misses Coverage

Ln 160, Col 51 100% Windows (CRLF) UTF-8

Activate Windows
Go to Settings to activate Windows.

Fsm

```
coverage_report - Notepad
File Edit Format View Help
Coverage Report by instance with details
=====
=== Instance: /top/if_t
=== Design Unit: work.ALSU_if
=====
Toggle Coverage:
Enabled Coverage      Bins    Hits    Misses  Coverage
-----
Toggles              80      80      0    100.00%

=====Toggle Details=====
Toggle Coverage for instance /top/if_t --

Node      1H->0L    0L->1H  "Coverage"
-----
A[2-0]      1          1    100.00
B[2-0]      1          1    100.00
bypass_A    1          1    100.00
bypass_B    1          1    100.00
cin         1          1    100.00
clk         1          1    100.00
direction   1          1    100.00
leds[15-0]  1          1    100.00
opcode[2-0] 1          1    100.00
out[5-0]    1          1    100.00
red_op_A    1          1    100.00
red_op_B    1          1    100.00
rst         1          1    100.00
serial_in   1          1    100.00

Total Node Count    =    40
Toggled Node Count  =    40
Untoggled Node Count =    0

Toggle Coverage     =   100.00% (80 of 80 bins)

=====
=== Instance: /top/sr_t
=== Design Unit: work.shift_reg_if
=====
Toggle Coverage:
Enabled Coverage      Bins    Hits    Misses  Coverage
-----
Toggles              30      30      0    100.00%

=====Toggle Details=====
Toggle Coverage for instance /top/sr_t --

Node      1H->0L    0L->1H  "Coverage"
-----
datain[5-0] 1          1    100.00
dataout[5-0] 1          1    100.00
direction    1          1    100.00
mode         1          1    100.00
serial_in    1          1    100.00

Total Node Count    =    15
Toggled Node Count  =    15
Untoggled Node Count =    0

Toggle Coverage     =   100.00% (30 of 30 bins)

=====
=== Instance: /top/SR
=== Design Unit: work.shift_reg
=====
Branch Coverage:
Enabled Coverage      Bins    Hits    Misses  Coverage
-----
Branches              5        5      0    100.00%

=====Branch Details=====
Branch Coverage for instance /top/SR

Line      Item      Count      Source
-----
File shift_reg.sv
-----IF Branch-----
13         1         47255      count coming in to IF
13         1         23201
19         1         11515
```

```
coverage_report - Notepad
File Edit Format View Help
Branches                5      5      0  100.00%

=====Branch Details=====

Branch Coverage for instance /top/SR

Line      Item      Count      Source
----      -
File shift_reg.sv
-----IF Branch-----
13          1      47255      Count coming in to IF
13          1      23201
19          1      11515
21          1      12539
Branch totals: 3 hits of 3 branches = 100.00%

-----IF Branch-----
14          1      23201      Count coming in to IF
14          1      11481
16          1      11720
Branch totals: 2 hits of 2 branches = 100.00%

Statement Coverage:
Enabled Coverage      Bins      Hits      Misses      Coverage
-----
Statements            5          5          0  100.00%

=====Statement Details=====

Statement Coverage for instance /top/SR --

Line      Item      Count      Source
----      -
File shift_reg.sv
11          1          1
15          1      11481
17          1      11720
20          1      11515
22          1      12539

=====
=== Instance: /top/DUT/AS
```

```
coverage_report - Notepad
File Edit Format View Help
=====
=== Instance: /top/DUT
=== Design Unit: work.ALSU
=====

Branch Coverage:
Enabled Coverage      Bins      Hits      Misses      Coverage
-----
Branches              28      28          0  100.00%

=====Branch Details=====

Branch Coverage for instance /top/DUT

Line      Item      Count      Source
----      -
File ALSU.sv
-----IF Branch-----
28          1      50948      Count coming in to IF
28          1      1937
40          1      49011
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
57          1      50948      Count coming in to IF
57          1      1937
59          1      49011
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
60          1      49011      Count coming in to IF
60          1      8696
62          1      40315
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
70          1      50948      Count coming in to IF
70          1      1937
73          1      49011
Branch totals: 2 hits of 2 branches = 100.00%

-----IF Branch-----
74          1      49011      Count coming in to IF
74          1      176

=====
Ln 160, Col 51  100%  Windows (CRLF)  UTF-8
```

coverage_report - Notepad

File Edit Format View Help

-----IF Branch-----

7449011Count coming in to IF

741176

7611560

7811566

8017123

82138586

Branch totals: 5 hits of 5 branches = 100.00%

-----CASE Branch-----

8338586Count coming in to CASE

8418604

9417650

10415535

11215478

11315674

11615644

11911

Branch totals: 7 hits of 7 branches = 100.00%

-----IF Branch-----

858604Count coming in to IF

851367

871222

891285

9117730

Branch totals: 4 hits of 4 branches = 100.00%

-----IF Branch-----

957650Count coming in to IF

951368

971246

991284

10116752

Branch totals: 4 hits of 4 branches = 100.00%

Condition Coverage:

Enabled Coverage	Bins	Covered	Misses	Coverage
Conditions	6	6	0	100.00%

Ln 358, Col 54100%Windows (CRLF)UTF-8

coverage_report - Notepad

File Edit Format View Help

=====Condition Details=====

Condition Coverage for instance /top/DUT --

File ALSU.sv

-----Focused Condition View-----

Line74Item1(bypass_A_reg && bypass_B_reg)

Condition totals: 2 of 2 input terms covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
bypass_A_reg	Y		
bypass_B_reg	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
Row 1:	1	bypass_A_reg_0	-
Row 2:	1	bypass_A_reg_1	bypass_B_reg
Row 3:	1	bypass_B_reg_0	bypass_A_reg
Row 4:	1	bypass_B_reg_1	bypass_A_reg

-----Focused Condition View-----

Line85Item1(red_op_A_reg && red_op_B_reg)

Condition totals: 2 of 2 input terms covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
red_op_A_reg	Y		
red_op_B_reg	Y		

Rows:	Hits	FEC Target	Non-masking condition(s)
Row 1:	1	red_op_A_reg_0	-
Row 2:	1	red_op_A_reg_1	red_op_B_reg
Row 3:	1	red_op_B_reg_0	red_op_A_reg
Row 4:	1	red_op_B_reg_1	red_op_A_reg

-----Focused Condition View-----

Line95Item1(red_op_A_reg && red_op_B_reg)

Condition totals: 2 of 2 input terms covered = 100.00%

Input Term	Covered	Reason for no coverage	Hint
------------	---------	------------------------	------

Ln 396, Col 69100%Windows (CRLF)UTF-8


```
coverage_report - Notepad
File Edit Format View Help
Statement Coverage:
Enabled Coverage
-----
Statements      Bins      Hits      Misses      Coverage
-----
Statements      50        50         0      100.00%

=====Statement Details=====

Statement Coverage for instance /top/DUT --

Line      Item      Count      Source
----      -
File ALSU.sv
13         1         47833
14         1         45634
15         1         9886
23         1         37668
24         1         40560
27         1         50948
29         1         1937
30         1         1937
31         1         1937
32         1         1937
33         1         1937
34         1         1937
35         1         1937
36         1         1937
37         1         1937
38         1         1937
41         1         49011
42         1         49011
43         1         49011
44         1         49011
45         1         49011
46         1         49011
47         1         49011
48         1         49011
49         1         49011
50         1         49011
56         1         50948
58         1         1937
61         1         8696
63         1         40315
68         1         50948

Ln 396, Col 69      100%      Windows (CRLF)      UTF-8
```

```
coverage_report - Notepad
File Edit Format View Help
68         1         50948
71         1         1937
75         1         176
77         1         1560
79         1         1566
81         1         7123
86         1         367
88         1         222
90         1         285
92         1         7730
96         1         368
98         1         246
100        1         284
102        1         6752
106        1         5535
112        1         5478
114        1         5674
117        1         5644
119        1         1
123        1         50948

Toggle Coverage:
Enabled Coverage
-----
Toggles      Bins      Hits      Misses      Coverage
-----
Toggles      64        64         0      100.00%

=====Toggle Details=====

Toggle Coverage for instance /top/DUT --

Node      1H->0L      0L->1H      "Coverage"
-----
A_reg[2-0]      1          1      100.00
B_reg[2-0]      1          1      100.00
bypass_A_reg      1          1      100.00
bypass_B_reg      1          1      100.00
cin_reg[1-0]      1          1      100.00
direction_reg      1          1      100.00
invalid      1          1      100.00
invalid_opcode      1          1      100.00
invalid_red_op      1          1      100.00
opcode_reg[2-0]      1          1      100.00
out_next[5-0]      1          1      100.00
out_shift_reg[5-0]      1          1      100.00

Ln 559, Col 51      100%      Windows (CRLF)      UTF-8
```

```
coverage_report - Notepad
File Edit Format View Help

invalid_red_op      1      1    100.00
opcode_reg[2-0]    1      1    100.00
out_next[5-0]      1      1    100.00
out_shift_reg[5-0] 1      1    100.00
red_op_A_reg       1      1    100.00
red_op_B_reg       1      1    100.00
serial_in_reg      1      1    100.00

Total Node Count    =      32
Toggled Node Count  =      32
Untoggled Node Count =       0

Toggle Coverage     =    100.00% (64 of 64 bins)

=====
=== Instance: /top
=== Design Unit: work.top
=====
Statement Coverage:
Enabled Coverage    Bins    Hits    Misses Coverage
-----
Statements          5      5      0    100.00%

=====Statement Details=====
Statement Coverage for instance /top --

Line    Item          Count    Source
----    -
File top.sv
7        1          100001
7        2          100000
17       1           1
18       1           1
19       1           1

Toggle Coverage:
Enabled Coverage    Bins    Hits    Misses Coverage
-----
Toggles            2      2      0    100.00%

=====Toggle Details=====
Toggle Coverage for instance /top --

Ln 601, Col 71    100%    Windows (CRLF)    UTF-8

coverage_report - Notepad
File Edit Format View Help
Statements          5      5      0    100.00%

=====Statement Details=====
Statement Coverage for instance /top --

Line    Item          Count    Source
----    -
File top.sv
7        1          100001
7        2          100000
17       1           1
18       1           1
19       1           1

Toggle Coverage:
Enabled Coverage    Bins    Hits    Misses Coverage
-----
Toggles            2      2      0    100.00%

=====Toggle Details=====
Toggle Coverage for instance /top --

Node      1H->0L    0L->1H    "Coverage"
-----
clk       1         1         100.00

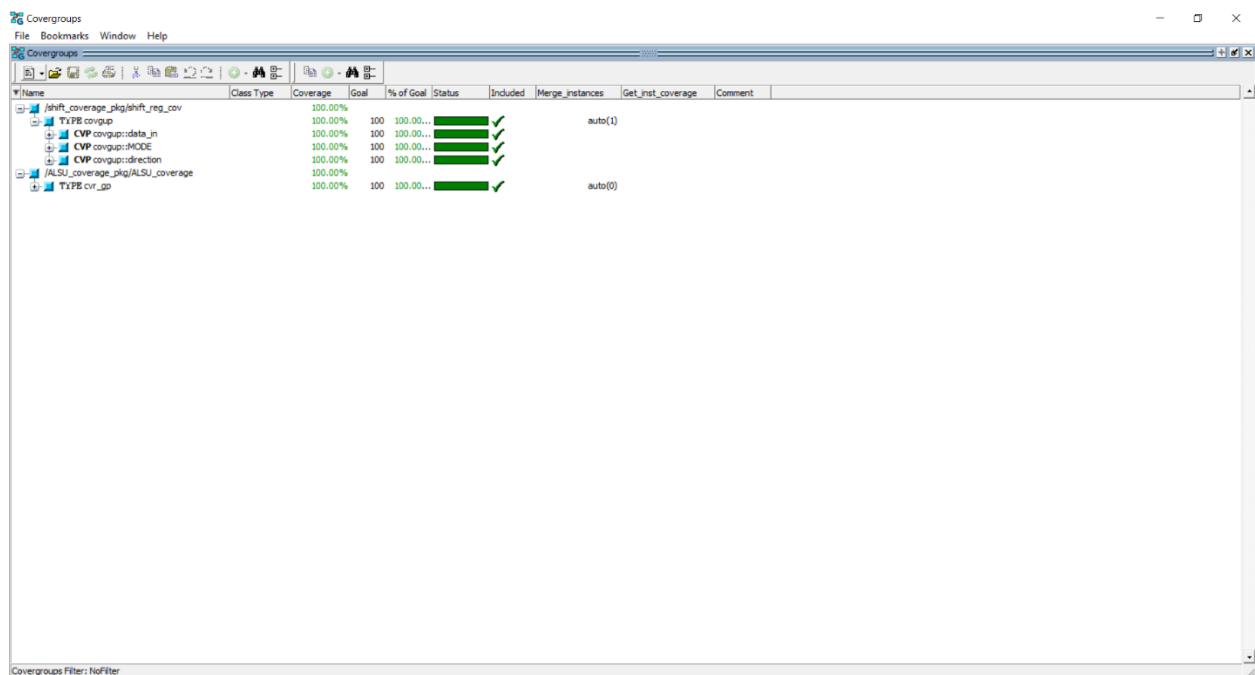
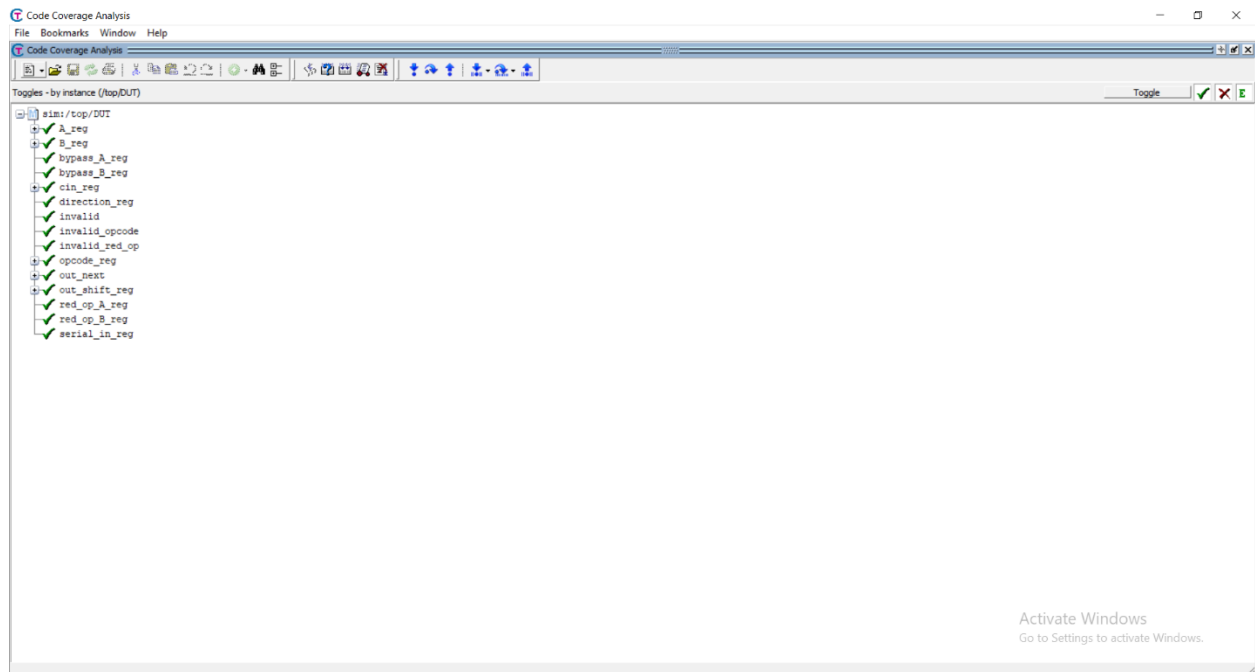
Total Node Count    =      1
Toggled Node Count  =      1
Untoggled Node Count =       0

Toggle Coverage     =    100.00% (2 of 2 bins)

=====
=== Instance: /config_object_pkg
=== Design Unit: work.config_object_pkg
=====
Branch Coverage:
Enabled Coverage    Bins    Hits    Misses Coverage
-----
Branches           10      0     10     0.00%

=====Branch Details=====

Ln 601, Col 71    100%    Windows (CRLF)    UTF-8
```



Code Coverage Analysis

File Bookmarks Window Help

Code Coverage Analysis

Statements - by instance (/top/DUT)

Statement

```
13 assign invalid_opcode = opcode_reg[1] & opcode_reg[0];
14 assign invalid = invalid_red_op | invalid_opcode;
23 if(if_t.rst) begin
24   cin_reg <= 0;
27   bypass_B_reg <= 0;
29   direction_reg <= 0;
30   serial_in_reg <= 0;
31   opcode_reg <= 0;
32   A_reg <= 0;
33   B_reg <= 0;
35 end else begin
36   cin_reg <= if_t.cin;
37   red_op_B_reg <= if_t.red_op_B;
38   red_op_A_reg <= if_t.red_op_A;
41   direction_reg <= if_t.direction;
42   serial_in_reg <= if_t.serial_in;
43   opcode_reg <= if_t.opcode;
44   A_reg <= if_t.A;
45   B_reg <= if_t.B;
47 end
48 end
50 always @(posedge if_t.clk or posedge if_t.rst) begin
56 else
58 end
61 always @(posedge if_t.clk or posedge if_t.rst) begin
63 if(if_t.rst) begin
68   if_t.out <= (if_t.INPUT_PRIORITY == "A")?A_reg: B_reg;
71 else if (bypass_B_reg)
75 else begin
77   begin
79   if_t.out <= (if_t.INPUT_PRIORITY == "A")? A_reg: B_reg;
81   if_t.out <= A_reg;
86 end
88 if (red_op_A_reg && red_op_B_reg)
90 else if (red_op_A_reg)
92 else if (red_op_B_reg)
96 end
```

Code Coverage Analysis

File Bookmarks Window Help

Code Coverage Analysis

Branches - by instance (/top/DUT)

Branch

```
28 bypass_A_reg <= 0;
40 bypass_A_reg <= if_t.bypass_A;
57 if_t.leds <= 0;
59 end
70 if_t.out <= A_reg;
73 else if (invalid)
74 if_t.out <= 0;
76 case (opcode_reg)
78 if (red_op_A_reg && red_op_B_reg)
80 else if (red_op_A_reg)
82 else if (red_op_B_reg)
84 else
85 if_t.out <= A_reg | B_reg;
87 begin
89 if_t.out <= (if_t.INPUT_PRIORITY == "A")? A_reg: B_reg;
91 if_t.out <= A_reg;
94 else
95 if_t.out <= A_reg ^ B_reg;
97 begin
99 if_t.out <= A_reg + B_reg+cin_reg;
101 if_t.out <= A_reg + B_reg;
104 end
112 default: if_t.out<=if_t.out;
113 endcase
116 out_next<=if_t.out;
119 endmodule
<NULL>
top.sv
```

coverage_report - Notepad

File Edit Format View Help

```

=====
=== Instance: /shift_coverage_pkg
=== Design Unit: work.shift_coverage_pkg
=====

```

Covergroup Coverage:

Covergroups	1	na	na	100.00%
Coverpoints/Crosses	3	na	na	
Covergroup Bins	68	68	0	100.00%

Covergroup	Metric	Goal	Bins	Status
TYPE /shift_coverage_pkg/shift_reg_cov/covgup	100.00%	100	-	Covered
covered/total bins:	68	68	-	
missing/total bins:	0	68	-	
% Hit:	100.00%	100	-	
coverpoint data_in	100.00%	100	-	Covered
covered/total bins:	64	64	-	
missing/total bins:	0	64	-	
% Hit:	100.00%	100	-	
bin auto[0]	24403	1	-	Covered
bin auto[1]	7150	1	-	Covered
bin auto[2]	4225	1	-	Covered
bin auto[3]	5207	1	-	Covered
bin auto[4]	1326	1	-	Covered
bin auto[5]	461	1	-	Covered
bin auto[6]	1318	1	-	Covered
bin auto[7]	174	1	-	Covered
bin auto[8]	429	1	-	Covered
bin auto[9]	577	1	-	Covered
bin auto[10]	53	1	-	Covered
bin auto[11]	6	1	-	Covered
bin auto[12]	423	1	-	Covered
bin auto[13]	54	1	-	Covered
bin auto[14]	56	1	-	Covered
bin auto[15]	152	1	-	Covered
bin auto[16]	326	1	-	Covered
bin auto[17]	32	1	-	Covered
bin auto[18]	57	1	-	Covered
bin auto[19]	32	1	-	Covered
bin auto[20]	5	1	-	Covered

Ln 601, Col 71 100% Windows (CRLF) UTF-8

coverage_report - Notepad

File Edit Format View Help

bin auto[20]	5	1	-	Covered
bin auto[21]	3	1	-	Covered
bin auto[22]	1	1	-	Covered
bin auto[23]	6	1	-	Covered
bin auto[24]	79	1	-	Covered
bin auto[25]	23	1	-	Covered
bin auto[26]	48	1	-	Covered
bin auto[27]	18	1	-	Covered
bin auto[28]	86	1	-	Covered
bin auto[29]	242	1	-	Covered
bin auto[30]	640	1	-	Covered
bin auto[31]	1089	1	-	Covered
bin auto[32]	996	1	-	Covered
bin auto[33]	712	1	-	Covered
bin auto[34]	71	1	-	Covered
bin auto[35]	55	1	-	Covered
bin auto[36]	68	1	-	Covered
bin auto[37]	13	1	-	Covered
bin auto[38]	26	1	-	Covered
bin auto[39]	27	1	-	Covered
bin auto[40]	25	1	-	Covered
bin auto[41]	50	1	-	Covered
bin auto[42]	8	1	-	Covered
bin auto[43]	16	1	-	Covered
bin auto[44]	4	1	-	Covered
bin auto[45]	25	1	-	Covered
bin auto[46]	75	1	-	Covered
bin auto[47]	267	1	-	Covered
bin auto[48]	154	1	-	Covered
bin auto[49]	108	1	-	Covered
bin auto[50]	49	1	-	Covered
bin auto[51]	99	1	-	Covered
bin auto[52]	337	1	-	Covered
bin auto[53]	199	1	-	Covered
bin auto[54]	80	1	-	Covered
bin auto[55]	523	1	-	Covered
bin auto[56]	635	1	-	Covered
bin auto[57]	745	1	-	Covered
bin auto[58]	1305	1	-	Covered
bin auto[59]	1272	1	-	Covered
bin auto[60]	4091	1	-	Covered
bin auto[61]	5825	1	-	Covered
bin auto[62]	6202	1	-	Covered

Ln 1105, Col 32 100% Windows (CRLF) UTF-8

```
coverage_report - Notepad
File Edit Format View Help
bin auto[62] 6202 1 - Covered
bin auto[63] 10568 1 - Covered
Coverpoint MODE 100.00% 100 - Covered
covered/total bins: 2 2 -
missing/total bins: 0 2 -
% Hit: 100.00% 100 -
bin mode0 42886 1 - Covered
bin mode1 40447 1 - Covered
Coverpoint direction 100.00% 100 - Covered
covered/total bins: 2 2 -
missing/total bins: 0 2 -
% Hit: 100.00% 100 -
bin direction0 43349 1 - Covered
bin direction1 39984 1 - Covered
Statement Coverage:
Enabled Coverage Bins Hits Misses Coverage
-----
Statements 14 12 2 85.71%

=====Statement Details=====
Statement Coverage for instance /shift_coverage_pkg --
Line Item Count Source
----
File shift_coverage.sv
8 1 ***g***
8 2 ***g***
8 3 2
28 1 1
29 1 1
33 1 1
34 1 1
35 1 1
39 1 1
40 1 1
44 1 1
45 1 1
46 1 83334
48 1 83333

=====
=== Instance: /shift_scoreboard_pkg

Ln 1147, Col 42 100% Windows (CRLF) UTF-8

coverage_report - Notepad
File Edit Format View Help
missing/total bins: 0 6 -
% Hit: 100.00% 100 -
Coverpoint CB1 100.00% 100 - Covered
covered/total bins: 5 5 -
missing/total bins: 0 5 -
% Hit: 100.00% 100 -
Coverpoint CB2 100.00% 100 - Covered
covered/total bins: 5 5 -
missing/total bins: 0 5 -
% Hit: 100.00% 100 -
Coverpoint A_M 100.00% 100 - Covered
covered/total bins: 2 2 -
missing/total bins: 0 2 -
% Hit: 100.00% 100 -
Coverpoint CB3 100.00% 100 - Covered
covered/total bins: 7 7 -
missing/total bins: 0 7 -
% Hit: 100.00% 100 -
Cross corner_case 100.00% 100 - Covered
covered/total bins: 18 18 -
missing/total bins: 0 18 -
% Hit: 100.00% 100 -
Cross Addition 100.00% 100 - Covered
covered/total bins: 2 2 -
missing/total bins: 0 2 -
% Hit: 100.00% 100 -
Cross shift 100.00% 100 - Covered
covered/total bins: 2 2 -
missing/total bins: 0 2 -
% Hit: 100.00% 100 -
Cross shift_rotate 100.00% 100 - Covered
covered/total bins: 2 2 -
missing/total bins: 0 2 -
% Hit: 100.00% 100 -
Cross walkingones 100.00% 100 - Covered
covered/total bins: 2 2 -
missing/total bins: 0 2 -
% Hit: 100.00% 100 -
Cross invalidation 100.00% 100 - Covered
covered/total bins: 2 2 -
missing/total bins: 0 2 -
% Hit: 100.00% 100 -
Covergroup instance \ALSU_coverage_pkg::ALSU_coverage: cvr_gp

Ln 1147, Col 42 100% Windows (CRLF) UTF-8
```

coverage_report - Notepad				
File	Edit	Format	View	Help
% Hit:	100.00%	100	-	
Cross Addition	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Cross shift	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Cross shift rotate	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Cross walkingones	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Cross invalidation	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Covergroup instance \\\ALSU_coverage_pkg::ALSU_coverage::cvt_gp	100.00%	100	-	Covered
covered/total bins:	63	67	-	
missing/total bins:	4	67	-	
% Hit:	94.02%	100	-	
Coverpoint CIN [1]	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
bin CIN0	25143	1	-	Covered
bin CIN1	24857	1	-	Covered
Coverpoint red_op_A [1]	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
bin red_op_A0	42007	1	-	Covered
bin red_op_A1	7993	1	-	Covered
Coverpoint red_op_B [1]	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
bin red_op_B0	41786	1	-	Covered

Ln 1709, Col 44 100% Windows (CRLF) UTF-8

Activate Windows
Go to Settings to activate Windows.

coverage_report - Notepad				
File	Edit	Format	View	Help
bin direction0	25342	1	-	Covered
bin direction1	24658	1	-	Covered
Coverpoint serial_in [1]	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
bin serial_in0	24998	1	-	Covered
bin serial_in1	25002	1	-	Covered
Coverpoint special [1]	100.00%	100	-	Covered
covered/total bins:	6	6	-	
missing/total bins:	0	6	-	
% Hit:	100.00%	100	-	
bin operations[0]	8589	1	-	Covered
bin operations[1]	8588	1	-	Covered
bin operations[2]	7720	1	-	Covered
bin operations[3]	7660	1	-	Covered
bin operations[4]	8639	1	-	Covered
bin operations[5]	8725	1	-	Covered
Coverpoint CB1	100.00%	100	-	Covered
covered/total bins:	5	5	-	
missing/total bins:	0	5	-	
% Hit:	100.00%	100	-	
bin A_data_0	8210	1	-	Covered
bin A_data_max	7616	1	-	Covered
bin A_data_min	6873	1	-	Covered
bin A_data_walkingones[1]	581	1	-	Covered
bin A_data_walkingones[2]	604	1	-	Covered
default bin A_data_default	16292	-	-	Occurred
Coverpoint CB2	100.00%	100	-	Covered
covered/total bins:	5	5	-	
missing/total bins:	0	5	-	
% Hit:	100.00%	100	-	
bin B_data_0	9111	1	-	Covered
bin B_data_max	7576	1	-	Covered
bin B_data_min	5739	1	-	Covered
bin B_data_walkingones[1]	586	1	-	Covered
bin B_data_walkingones[2]	567	1	-	Covered
default bin B_data_default	16654	-	-	Occurred
Coverpoint A_M	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
bin Bins_arith[2]	7720	1	-	Covered

Ln 1709, Col 44 100% Windows (CRLF) UTF-8

Activate Windows
Go to Settings to activate Windows.

coverage_report - Notepad				
File Edit Format View Help				
bin A_data_min	6873	1	-	Covered
bin A_data_walkingones[1]	581	1	-	Covered
bin A_data_walkingones[2]	604	1	-	Covered
default bin A_data_default	16292		-	Occurred
Coverpoint CB2	100.00%	100	-	Covered
covered/total bins:	5	5	-	
missing/total bins:	0	5	-	
% Hit:	100.00%	100	-	
bin B_data_0	9111	1	-	Covered
bin B_data_max	7576	1	-	Covered
bin B_data_min	5739	1	-	Covered
bin B_data_walkingones[1]	586	1	-	Covered
bin B_data_walkingones[2]	567	1	-	Covered
default bin B_data_default	16654		-	Occurred
Coverpoint A_M	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
bin Bins_arith[2]	7720	1	-	Covered
bin Bins_arith[3]	7660	1	-	Covered
Coverpoint CB3	100.00%	100	-	Covered
covered/total bins:	7	7	-	
missing/total bins:	0	7	-	
% Hit:	100.00%	100	-	
illegal_bin Bins_invalid	79		-	Occurred
bin Bins_shift[4]	8639	1	-	Covered
bin Bins_shift[5]	8725	1	-	Covered
bin Bins_arith[2]	7720	1	-	Covered
bin Bins_arith[3]	7660	1	-	Covered
bin Bins_bitwise[0]	8589	1	-	Covered
bin Bins_bitwise[1]	8588	1	-	Covered
bin Bins_trans	10	1	-	Covered
Cross_corner_case	100.00%	100	-	Covered
covered/total bins:	18	18	-	
missing/total bins:	0	18	-	
% Hit:	100.00%	100	-	
Auto, Default and User Defined Bins:				
bin <Bins_arith[3],0_data_min,A_data_min>				
	95	1	-	Covered
bin <Bins_arith[2],0_data_min,A_data_min>				
	94	1	-	Covered
bin <Bins_arith[3],0_data_min,A_data_max>				
	107	1	-	Covered

Activate Windows
Go to Settings to activate Windows.

Ln 1709, Col 44 100% Windows (CRLF) UTF-8

coverage_report - Notepad				
File Edit Format View Help				
Auto, Default and User Defined Bins:				
bin <Bins_arith[3],0_data_min,A_data_min>	95	1	-	Covered
bin <Bins_arith[2],0_data_min,A_data_min>	94	1	-	Covered
bin <Bins_arith[3],0_data_min,A_data_max>				
	107	1	-	Covered
bin <Bins_arith[3],0_data_min,A_data_0>				
	104	1	-	Covered
bin <Bins_arith[2],0_data_min,A_data_max>				
	111	1	-	Covered
bin <Bins_arith[2],0_data_min,A_data_0>				
	101	1	-	Covered
bin <Bins_arith[3],0_data_max,A_data_min>				
	119	1	-	Covered
bin <Bins_arith[3],0_data_0,A_data_min>				
	124	1	-	Covered
bin <Bins_arith[2],0_data_max,A_data_min>				
	110	1	-	Covered
bin <Bins_arith[2],0_data_0,A_data_min>				
	130	1	-	Covered
bin <Bins_arith[3],0_data_max,A_data_max>				
	939	1	-	Covered
bin <Bins_arith[3],0_data_0,A_data_max>				
	116	1	-	Covered
bin <Bins_arith[3],0_data_max,A_data_0>				
	144	1	-	Covered
bin <Bins_arith[3],0_data_0,A_data_0>				
	950	1	-	Covered
bin <Bins_arith[2],0_data_max,A_data_max>				
	970	1	-	Covered
bin <Bins_arith[2],0_data_0,A_data_max>				
	109	1	-	Covered
bin <Bins_arith[2],0_data_max,A_data_0>				
	140	1	-	Covered
bin <Bins_arith[2],0_data_0,A_data_0>				
	994	1	-	Covered
Illegal and Ignore Bins:				
ignore_bin walkingB	129		-	Occurred
ignore_bin walkingA	108		-	Occurred
Cross Addition	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Auto, Default and User Defined Bins:				

Activate Windows
Go to Settings to activate Windows.

Ln 1813, Col 55 100% Windows (CRLF) UTF-8

coverage_report - Notepad				
File	Edit	Format	View	Help
bin <Bins_arith[2],0_data_0,A_data_max>				
	109	1	-	Covered
bin <Bins_arith[2],0_data_max,A_data_0>				
	140	1	-	Covered
bin <Bins_arith[2],0_data_0,A_data_0>				
	994	1	-	Covered
Illegal and Ignore Bins:				
ignore_bin walkingB	129		-	Occurred
ignore_bin walkingA	108		-	Occurred
Cross Addition				
	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Auto, Default and User Defined Bins:				
bin Add_cin0	3867	1	-	Covered
bin Add_cin1	3853	1	-	Covered
Cross shift				
	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Auto, Default and User Defined Bins:				
bin shift_si0	4340	1	-	Covered
bin shift_si1	4340	1	-	Covered
Cross shift_rotate				
	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Auto, Default and User Defined Bins:				
bin shu_rot_d0	8846	1	-	Covered
bin shu_rot_d1	8518	1	-	Covered
Cross walkingones				
	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Auto, Default and User Defined Bins:				
bin arithA	104	1	-	Covered
bin arithB	47	1	-	Covered
Cross invalidation				
	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Auto, Default and User Defined Bins:				
bin R0pA_notXor	6314	1	-	Covered
bin R0pB_notXor	6422	1	-	Covered

Activate Windows
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Ln 1846, Col 67 100% Windows (CRLF) UTF-8

coverage_report - Notepad				
File	Edit	Format	View	Help
bin auto[1]				
	7150	1	-	Covered
bin auto[2]				
	4225	1	-	Covered
bin auto[3]				
	5207	1	-	Covered
bin auto[4]				
	1326	1	-	Covered
bin auto[5]				
	461	1	-	Covered
bin auto[6]				
	1318	1	-	Covered
bin auto[7]				
	174	1	-	Covered
bin auto[8]				
	429	1	-	Covered
bin auto[9]				
	577	1	-	Covered
bin auto[10]				
	53	1	-	Covered
bin auto[11]				
	6	1	-	Covered
bin auto[12]				
	423	1	-	Covered
bin auto[13]				
	54	1	-	Covered
bin auto[14]				
	56	1	-	Covered
bin auto[15]				
	152	1	-	Covered
bin auto[16]				
	326	1	-	Covered
bin auto[17]				
	32	1	-	Covered
bin auto[18]				
	57	1	-	Covered
bin auto[19]				
	32	1	-	Covered
bin auto[20]				
	5	1	-	Covered
bin auto[21]				
	3	1	-	Covered
bin auto[22]				
	1	1	-	Covered
bin auto[23]				
	6	1	-	Covered
bin auto[24]				
	79	1	-	Covered
bin auto[25]				
	23	1	-	Covered
bin auto[26]				
	48	1	-	Covered
bin auto[27]				
	18	1	-	Covered
bin auto[28]				
	86	1	-	Covered
bin auto[29]				
	242	1	-	Covered
bin auto[30]				
	640	1	-	Covered
bin auto[31]				
	1089	1	-	Covered
bin auto[32]				
	996	1	-	Covered
bin auto[33]				
	712	1	-	Covered
bin auto[34]				
	71	1	-	Covered
bin auto[35]				
	55	1	-	Covered
bin auto[36]				
	68	1	-	Covered
bin auto[37]				
	13	1	-	Covered
bin auto[38]				
	26	1	-	Covered
bin auto[39]				
	27	1	-	Covered
bin auto[40]				
	25	1	-	Covered
bin auto[41]				
	50	1	-	Covered
bin auto[42]				
	8	1	-	Covered
bin auto[43]				
	16	1	-	Covered

Activate Windows
Go to Settings to activate Windows.

Ln 1846, Col 67 100% Windows (CRLF) UTF-8

coverage_report - Notepad				
File Edit Format View Help				
% Hit:	100.00%	100	-	
Coverpoint serial_in	0.00%	100	-	ZERO
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Coverpoint special	0.00%	100	-	ZERO
covered/total bins:	6	6	-	
missing/total bins:	0	6	-	
% Hit:	100.00%	100	-	
Coverpoint CB1	100.00%	100	-	Covered
covered/total bins:	5	5	-	
missing/total bins:	0	5	-	
% Hit:	100.00%	100	-	
Coverpoint CB2	100.00%	100	-	Covered
covered/total bins:	5	5	-	
missing/total bins:	0	5	-	
% Hit:	100.00%	100	-	
Coverpoint A_M	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Coverpoint CB3	100.00%	100	-	Covered
covered/total bins:	7	7	-	
missing/total bins:	0	7	-	
% Hit:	100.00%	100	-	
Cross corner_case	100.00%	100	-	Covered
covered/total bins:	18	18	-	
missing/total bins:	0	18	-	
% Hit:	100.00%	100	-	
Cross Adddation	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Cross shift	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Cross shift_rotate	100.00%	100	-	Covered
covered/total bins:	2	2	-	
missing/total bins:	0	2	-	
% Hit:	100.00%	100	-	
Cross walkingones	100.00%	100	-	Covered
covered/total bins:	2	2	-	

Activate Windows
Go to Settings to activate Windows.